

A Living Review of Quantum Information Science in High Energy Physics

Pamela Pajarillo, So Chigusa, Sokratis Trifinopoulos, Jesse Thaler

*Center for Theoretical Physics - a Leinweber Institute, Massachusetts Institute of Technology
Cambridge, MA 02139, USA*

E-mail: pampaja@mit.edu, schigusa@mit.edu, trifinos@mit.edu,
jthaler@mit.edu

ABSTRACT: Inspired by “A Living Review of Machine Learning for Particle Physics”¹, the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

¹See <https://github.com/iml-wg/HEPML-LivingReview>.

Contents

1	Introduction	1
2	Nuclear and Particle Physics in Quantum Information Science	2
2.1	Reviews and Whitepapers and Proceedings	2
2.1.1	Reviews	2
2.1.2	Whitepapers and Proceedings	2
2.2	Anomaly Detection	6
2.2.1	Quantum Kernel Methods	6
2.2.2	Quantum k-Means	7
2.2.3	Variational Quantum Algorithms	7
2.2.4	Crosslists	8
2.3	Beyond the Standard Model	8
2.3.1	Quantum Algorithms - Grover's Search Algorithm	8
2.3.2	Crosslists	8
2.4	Dark Matter - Particle-like Dark Matter	9
2.4.1	Quantum Sensors - Atomic Sensors	9
2.4.2	Quantum Sensors - Optomechanical Sensors	10
2.4.3	Quantum Sensors - Solid State Sensors	12
2.4.4	Quantum Simulations	13
2.4.5	Crosslists	13
2.5	Dark Matter - Wave-like Dark Matter	15
2.5.1	Quantum Sensors - Atomic Sensors	15
2.5.2	Quantum Sensors - Nuclear Sensors	16
2.5.3	Quantum Sensors - Optical/Photonic Sensors	16
2.5.4	Quantum Sensors - Optomechanical Sensors	19
2.5.5	Quantum Sensors - Solid State Sensors	20
2.5.6	Crosslists	22
2.6	Dark Matter - Macroscopic Dark Matter	25
2.6.1	Quantum Sensors - Optomechanical Sensors	25
2.6.2	Crosslists	25
2.7	Detector Technologies and Simulations	26
2.7.1	Continuous Variable Quantum Computing	26
2.7.2	Quantum Annealing	26
2.7.3	Quantum Generative Adversarial Networks	27
2.7.4	Quantum Sensors - Atomic Sensors	27
2.7.5	Crosslists	28
2.8	Effective Field Theories	28

2.8.1	Quantum Simulations	28
2.8.2	Crosslists	29
2.9	Event Classification	29
2.9.1	Quantum Annealing	29
2.9.2	Quantum Kernel Methods	30
2.9.3	Quantum Neural Networks	30
2.9.4	Variational Quantum Algorithms	30
2.9.5	Crosslists	32
2.10	Event Generation	34
2.10.1	Quantum Algorithms - Grover's Search Algorithm	34
2.10.2	Quantum Algorithms - Quantum Walks	34
2.10.3	Quantum Generative Adversarial Networks	35
2.10.4	Quantum Neural Networks	35
2.10.5	Quantum Simulations	35
2.10.6	Variational Quantum Algorithms	36
2.10.7	Crosslists	36
2.11	Fundamental Symmetries and Interactions	38
2.11.1	Quantum Entanglement and Bell Inequalities	38
2.11.2	Crosslists	38
2.12	Gravitation and Astrophysics	39
2.12.1	Quantum Annealing	39
2.12.2	Quantum Sensors - Optical/Photonic Sensors	39
2.12.3	Quantum Sensors - Optomechanical Sensors	39
2.12.4	Crosslists	40
2.13	Higgs Physics	43
2.13.1	Quantum Algorithms - Grover's Search Algorithm	43
2.13.2	Quantum Annealing	43
2.13.3	Quantum Kernel Methods	44
2.13.4	Variational Quantum Algorithms	44
2.13.5	Crosslists	44
2.14	Jet Reconstruction	45
2.14.1	Quantum Algorithms - Grover's Search Algorithm	45
2.14.2	Quantum Annealing	45
2.14.3	Quantum k-Means	46
2.14.4	Tensor Networks	46
2.14.5	Variational Quantum Algorithms	46
2.14.6	Crosslists	47
2.15	Lattice Scalar Theories	47
2.15.1	Quantum Simulations	47
2.16	Lattice Fermion Theories	48
2.16.1	Quantum Simulations	48

2.17	Lattice Gauge Theories	48
2.17.1	Quantum Annealing	48
2.17.2	Quantum Neural Networks	48
2.17.3	Quantum Simulations	49
2.17.4	Variational Quantum Algorithms	52
2.17.5	Crosslists	52
2.18	Neutrinos	55
2.18.1	Quantum Sensors - Optomechanical Sensors	55
2.18.2	Variational Quantum Algorithms	55
2.18.3	Crosslists	55
2.19	Nuclear Decays	56
2.19.1	Quantum Sensors - Optomechanical Sensors	56
2.20	Quantum Chromodynamics	56
2.20.1	Quantum Entanglement and Bell Inequalities	56
2.20.2	Quantum Simulations	56
2.20.3	Crosslists	57
2.21	Quantum Gravity	57
2.21.1	Crosslists	57
2.22	Quantum Field Theories	59
2.22.1	Quantum Entanglement and Bell Inequalities	59
2.22.2	Crosslists	59
2.23	Spin Correlation in Particle Decays	60
2.23.1	Quantum Entanglement and Bell Inequalities	60
2.24	Statistical Analysis	63
2.24.1	Quantum Annealing	63
2.25	Schwinger Model	64
2.25.1	Quantum Simulations	64
2.25.2	Variational Quantum Algorithms	64
2.26	Track Reconstruction	65
2.26.1	Quantum Algorithms - Grover's Search Algorithm	65
2.26.2	Quantum Algorithms - Harrow-Hassadim-Lloyd Algorithm	65
2.26.3	Quantum Annealing	65
2.26.4	Quantum Kernel Methods	66
2.26.5	Quantum Memory	66
2.26.6	Quantum Neural Networks	67
2.26.7	Variational Quantum Algorithms	67
2.26.8	Crosslists	67

3	Quantum Information Science in Nuclear and Particle Physics	69
3.1	Reviews and Whitepapers and Proceedings	69
3.1.1	Reviews	69
3.1.2	Whitepapers and Proceedings	69
3.2	Continuous Variable Quantum Computing	73
3.2.1	Detector Technologies and Simulations	73
3.2.2	Crosslists	74
3.3	Quantum Algorithms - Grover's Search Algorithm	74
3.3.1	Beyond the Standard Model	74
3.3.2	Event Generation	74
3.3.3	Higgs Physics	74
3.3.4	Jet Reconstruction	74
3.3.5	Track Reconstruction	75
3.3.6	Crosslists	75
3.4	Quantum Algorithms - Harrow-Hassadim-Lloyd Algorithm	75
3.4.1	Track Reconstruction	75
3.4.2	Crosslists	75
3.5	Quantum Algorithms - Quantum Walks	76
3.5.1	Event Generation	76
3.6	Quantum Annealing	76
3.6.1	Detector Technologies and Simulations	76
3.6.2	Event Classification	76
3.6.3	Gravitation and Astrophysics	76
3.6.4	Higgs Physics	77
3.6.5	Jet Reconstruction	77
3.6.6	Lattice Gauge Theories	78
3.6.7	Statistical Analysis	78
3.6.8	Track Reconstruction	78
3.6.9	Crosslists	79
3.7	Quantum Entanglement and Bell Inequalities	81
3.7.1	Fundamental Symmetries and Interactions	81
3.7.2	Quantum Chromodynamics	81
3.7.3	Quantum Field Theories	81
3.7.4	Spin Correlation in Particle Decays	81
3.7.5	Crosslists	84
3.8	Quantum Generative Adversarial Networks	85
3.8.1	Detector Technologies and Simulations	85
3.8.2	Event Generation	85
3.8.3	Crosslists	86
3.9	Quantum Kernel Methods	87
3.9.1	Anomaly Detection	87

3.9.2	Event Classification	87
3.9.3	Higgs Physics	88
3.9.4	Track Reconstruction	88
3.9.5	Crosslists	88
3.10	Quantum k-Means	89
3.10.1	Anomaly Detection	89
3.10.2	Jet Reconstruction	89
3.10.3	Crosslists	90
3.11	Quantum Memory	90
3.11.1	Track Reconstruction	90
3.11.2	Crosslists	90
3.12	Quantum Neural Networks	90
3.12.1	Event Classification	90
3.12.2	Event Generation	91
3.12.3	Lattice Gauge Theories	91
3.12.4	Track Reconstruction	91
3.12.5	Crosslists	91
3.13	Quantum Sensors - Atomic Sensors	92
3.13.1	Dark Matter - Particle-like Dark Matter	92
3.13.2	Dark Matter - Wave-like Dark Matter	93
3.13.3	Detector Technologies and Simulations	94
3.13.4	Crosslists	94
3.14	Quantum Sensors - Molecular Sensors	97
3.14.1	Crosslists	97
3.15	Quantum Sensors - Nuclear Sensors	99
3.15.1	Dark Matter - Wave-like Dark Matter	99
3.15.2	Crosslists	100
3.16	Quantum Sensors - Optical/Photonic Sensors	102
3.16.1	Dark Matter - Wave-like Dark Matter	102
3.16.2	Gravitation and Astrophysics	104
3.16.3	Crosslists	105
3.17	Quantum Sensors - Optomechanical Sensors	108
3.17.1	Dark Matter - Particle-like Dark Matter	108
3.17.2	Dark Matter - Wave-like Dark Matter	110
3.17.3	Dark Matter - Macroscopic Dark Matter	112
3.17.4	Gravitation and Astrophysics	112
3.17.5	Neutrinos	112
3.17.6	Nuclear Decays	112
3.17.7	Crosslists	113
3.18	Quantum Sensors - Solid State Sensors	115
3.18.1	Dark Matter - Particle-like Dark Matter	115

3.18.2	Dark Matter - Wave-like Dark Matter	116
3.18.3	Crosslists	118
3.19	Quantum Simulations	121
3.19.1	Dark Matter - Particle-like Dark Matter	121
3.19.2	Effective Field Theories	122
3.19.3	Event Generation	122
3.19.4	Lattice Scalar Theories	122
3.19.5	Lattice Fermion Theories	123
3.19.6	Lattice Gauge Theories	123
3.19.7	Quantum Chromodynamics	126
3.19.8	Schwinger Model	127
3.19.9	Crosslists	127
3.20	Tensor Networks	129
3.20.1	Jet Reconstruction	129
3.20.2	Crosslists	130
3.21	Variational Quantum Algorithms	131
3.21.1	Anomaly Detection	131
3.21.2	Event Classification	131
3.21.3	Event Generation	133
3.21.4	Higgs Physics	133
3.21.5	Jet Reconstruction	133
3.21.6	Lattice Gauge Theories	134
3.21.7	Neutrinos	134
3.21.8	Schwinger Model	134
3.21.9	Track Reconstruction	135
3.21.10	Crosslists	135

Contents

1 Introduction

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. The papers are listed in reverse chronological order. In order to be as useful as possible, this document will continually change. Please check back ² regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

²See <https://github.com/PamelaPajarillo/NUPAQIS-LivingReview>.

2 Nuclear and Particle Physics in Quantum Information Science

2.1 Reviews and Whitepapers and Proceedings

2.1.1 Reviews

2.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics** **57 388 (2016):** 09 March 2016

2.1.1.2 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.1.1.3 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.1.1.4 Review on Quantum Computing for Lattice Field Theory [4]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

2.1.2 Whitepapers and Proceedings

2.1.2.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph

Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

2.1.2.2 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020

- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.1.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [7]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma

- **Posted on arXiv:** 09 March 2022

2.1.2.4 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle

- **Posted on arXiv:** 14 March 2022

2.1.2.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.1.2.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [10]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.1.2.7 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.1.2.8 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatriwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz,

S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

2.1.2.9 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.1.2.10 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022

- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.1.2.11 Quantum computing hardware for HEP algorithms and sensing [15]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canceledo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

2.1.2.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.1.2.13 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenktuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.1.2.14 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.2 Anomaly Detection

2.2.1 Quantum Kernel Methods

2.2.1.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [19]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.2.1.2 Quantum anomaly detection in the latent space of proton collision events at the LHC [20]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.2.2 Quantum k-Means

2.2.2.1 Unsupervised event classification with graphs on classical and photonic quantum computers [21]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.3 Variational Quantum Algorithms

2.2.3.1 Anomaly detection in high-energy physics using a quantum autoencoder [22]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.2.3.2 Quantum anomaly detection for collider physics [23]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

2.2.4 Crosslists

2.2.4.1 A Quantum Algorithm for Model-Independent Searches for New Physics [24]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.3 Beyond the Standard Model

2.3.1 Quantum Algorithms - Grover's Search Algorithm

2.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [25]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

2.3.2 Crosslists

2.3.2.1 Event Classification with Quantum Machine Learning in High-Energy Physics [26]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.3.2.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [27]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.3.2.3 Quantum algorithm for the classification of supersymmetric top quark events [28]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.3.2.4 Unravelling physics beyond the standard model with classical and quantum anomaly detection [19]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.3.2.5 Quantum anomaly detection in the latent space of proton collision events at the LHC [20]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.3.2.6 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.3.2.7 Quantum Metric Learning for New Physics Searches at the LHC [29]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.4 Dark Matter - Particle-like Dark Matter

2.4.1 Quantum Sensors - Atomic Sensors

2.4.1.1 Trapped Electrons and Ions as Particle Detectors [30]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor

- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

2.4.1.2 Millicharged Dark Matter Detection with Ion Traps [31]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

2.4.2 Quantum Sensors - Optomechanical Sensors

2.4.2.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [32]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

2.4.2.2 Proposal for gravitational direct detection of dark matter [33]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

2.4.2.3 Search for composite dark matter with optically levitated sensors [34]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

2.4.2.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [35]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

2.4.2.5 Requirements on quantum superpositions of macro-objects for sensing neutrinos [36]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

2.4.2.6 Optomechanical dark matter instrument for direct detection [37]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

2.4.2.7 Superfluid helium ultralight dark matter detector [38]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

2.4.2.8 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [39]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

2.4.2.9 Dark Matter Searches with Levitated Sensors [40]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

2.4.2.10 Probing Light Particles With Optically Trapped Sensors Through Nucleon Scattering [41]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeusun
- **Posted on arXiv:** 31 January 2025

2.4.2.11 Mechanical sensors for ultraheavy dark matter searches via long-range forces [42]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

2.4.3 Quantum Sensors - Solid State Sensors

2.4.3.1 Dark matter direct detection with quantum dots [43]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

2.4.3.2 Dark Matter Induced Power in Quantum Devices [44]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

2.4.3.3 Transmon Qubit constraints on dark matter-nucleon scattering [45]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

2.4.3.4 Quantum Parity Detectors: a qubit based particle detection scheme with meV thresholds for rare-event searches [46]

- **Authors:** Karthik Ramanathan, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Brandon J. Sandoval, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024

2.4.3.5 Piezoelectric bulk acoustic resonators for dark photon detection [47]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

2.4.4 Quantum Simulations

2.4.4.1 Quantum simulations of dark sector showers [48]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

2.4.5 Crosslists

2.4.5.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.4.5.2 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.4.5.3 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [49]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.4.5.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.4.5.5 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

2.4.5.6 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.5 Dark Matter - Wave-like Dark Matter

2.5.1 Quantum Sensors - Atomic Sensors

2.5.1.1 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [50]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

2.5.1.2 Quantum entanglement of ions for light dark matter detection [51]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

2.5.1.3 Highly excited electron cyclotron for QCD axion and dark-photon detection [52]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

2.5.1.4 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [53]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

2.5.1.5 Sensitively searching for microwave dark photons with atomic ensembles [54]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

2.5.1.6 Searches for exotic spin-dependent interactions with spin sensors [55]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

2.5.2 Quantum Sensors - Nuclear Sensors

2.5.2.1 Intensity interferometry for ultralight bosonic dark matter detection [56]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

2.5.2.2 Detection of bosonovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

2.5.2.3 A Nuclear Interferometer for Ultra-Light Dark Matter Detection [58]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024

2.5.2.4 Spin squeezing of macroscopic nuclear spin ensembles [59]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

2.5.3 Quantum Sensors - Optical/Photonic Sensors

2.5.3.1 Laser Interferometers as Dark Matter Detectors [60]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

2.5.3.2 Accelerating dark-matter axion searches with quantum measurement technology [61]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

2.5.3.3 Searching for Dark Matter with a Superconducting Qubit [62]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.5.3.4 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [63]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

2.5.3.5 Quantum measurements in fundamental physics: a user’s manual [64]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

2.5.3.6 Quantum Enhancement in Dark Matter Detection with Quantum Computation [65]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

2.5.3.7 Search for QCD axion dark matter with transmon qubits and quantum circuit [66]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

2.5.3.8 Axion bounds from quantum technology [67]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

2.5.3.9 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [68]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

2.5.3.10 Transmon qubit modeling and characterization for Dark Matter search [69]

- **Authors:** R. Moretti, D. Labranca, P. Campana, R. Carobene, M. Gobbo, M.A. Castellanos-Beltran, D. Olaya, P.F. Hopkins, L. Banchi, M. Borghesi, A. Candido, S. Carrazza, H.A. Corti, A. D’Elia, M. Faverzani, E. Ferri, A. Nucciotti, L. Origo, A. Pasquale, A.S. Piedjou Komnang, A. Rettaroli, S. Tocci, C. Gatti, A. Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in nan:** 09 September 2024

2.5.3.11 Flux-Tunable Cavity for Dark Matter Detection [70]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

2.5.3.12 Probing sub-eV dark photon, scalar and axion-like particle dark matters with transmon qubits [71]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Published in Phys.Dark Univ.:** 15 November 2025

2.5.4 Quantum Sensors - Optomechanical Sensors

2.5.4.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [72]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

2.5.4.2 Dark Matter Direct Detection with Accelerometers [73]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

2.5.4.3 Ultralight dark matter detection with mechanical quantum sensors [74]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

2.5.4.4 Searching for scalar dark matter with compact mechanical resonators [75]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

2.5.4.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [76]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

2.5.4.6 Searching for vector dark matter with an optomechanical accelerometer [77]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

2.5.4.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [78]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.:** 01 September 2023

2.5.4.8 Axion detection with optomechanical cavities [79]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B:** 27 June 2024

2.5.4.9 Vector wave dark matter and terrestrial quantum sensors [80]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

2.5.5 Quantum Sensors - Solid State Sensors

2.5.5.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [81]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

2.5.5.2 Axion search with quantum nondemolition detection of magnons [82]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

2.5.5.3 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [83]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research** 4, 023162 (2022): 31 May 2022

2.5.5.4 Ultimate precision limit of noise sensing and dark matter search [84]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

2.5.5.5 Light dark matter search with nitrogen-vacancy centers in diamonds [85]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

2.5.5.6 Axion detection via superfluid ^3He ferromagnetic phase and quantum measurement techniques [86]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

2.5.5.7 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [87]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D** 111 (2025) 075028: 01 April 2025

2.5.5.8 Listening for new physics with quantum acoustics [88]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohriti Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

2.5.5.9 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [89]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

2.5.5.10 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [90]

- **Authors:** Wei Chao, Yu Gao, Ming-jie Jin, Xiao-sheng Liu, Xi-lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in nan:** 2025

2.5.5.11 Hybrid-spin decoupling for noise-resilient DC quantum sensing [91]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

2.5.6 Crosslists

2.5.6.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph

Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

2.5.6.2 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [92]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.5.6.3 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.5.6.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

2.5.6.5 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

2.5.6.6 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.5.6.7 Electromagnetic cavities as mechanical bars for gravitational waves [93]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D’Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.5.6.8 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.5.6.9 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.5.6.10 Multimessenger Astronomy Beyond the Standard Model: New Window from Quantum Sensors [95]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025

2.6 Dark Matter - Macroscopic Dark Matter

2.6.1 Quantum Sensors - Optomechanical Sensors

2.6.1.1 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [49]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.6.2 Crosslists

2.6.2.1 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.6.2.2 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.7 Detector Technologies and Simulations

2.7.1 Continuous Variable Quantum Computing

2.7.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [96]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.7.2 Quantum Annealing

2.7.2.1 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [97]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

2.7.3 Quantum Generative Adversarial Networks

2.7.3.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [98]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

2.7.3.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [99]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in J.Phys.Conf.Ser.:** 2023

2.7.3.3 Precise image generation on current noisy quantum computing devices [100]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

2.7.4 Quantum Sensors - Atomic Sensors

2.7.4.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [101]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.7.4.2 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.7.5 Crosslists

2.7.5.1 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.7.5.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.7.5.3 Quantum-centric Supercomputing for Physics Research [102]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.8 Effective Field Theories

2.8.1 Quantum Simulations

2.8.1.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [103]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

2.8.2 Crosslists

2.8.2.1 Quantum SMEFT tomography: Top quark pair production at the LHC [104]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.8.2.2 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.8.2.3 Minimal entanglement and emergent symmetries in low-energy QCD [105]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.9 Event Classification

2.9.1 Quantum Annealing

2.9.1.1 Quantum algorithm for the classification of supersymmetric top quark events [28]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.9.1.2 Completely quantum neural networks [106]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

2.9.2 Quantum Kernel Methods

2.9.2.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [107]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevier
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

2.9.2.2 Application of quantum machine learning in a Higgs physics study at the CEPC [108]

- **Authors:** Abdualazem Fadol, Qiyu Sha, Yaquan Fang, Zhan Li, Sitian Qian, Yuyang Xiao, Yu Zhang, Chen Zhou
- **Posted on arXiv:** 26 September 2022
- **Published in International Journal of Modern Physics A, 2024:** 07 March 2024

2.9.3 Quantum Neural Networks

2.9.3.1 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [109]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

2.9.3.2 Guided graph compression for quantum graph neural networks [110]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

2.9.4 Variational Quantum Algorithms

2.9.4.1 Event Classification with Quantum Machine Learning in High-Energy Physics [26]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.9.4.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [27]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.9.4.3 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [111]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

2.9.4.4 Quantum Metric Learning for New Physics Searches at the LHC [29]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.9.4.5 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [112]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

2.9.4.6 Quantum Vision Transformers for Quark–Gluon Classification [113]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

2.9.5 Crosslists

2.9.5.1 Solving a Higgs optimization problem with quantum annealing for machine learning [114]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.9.5.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [115]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.9.5.3 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecora, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.9.5.4 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [116]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.9.5.5 Unsupervised event classification with graphs on classical and photonic quantum computers [21]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.9.5.6 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [117]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research** **3**, 033221 (2021): 08 September 2021

2.9.5.7 Higgs analysis with quantum classifiers [118]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.9.5.8 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [119]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.9.5.9 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.9.5.10 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.9.5.11 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.9.5.12 Quantum-centric Supercomputing for Physics Research [102]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.10 Event Generation

2.10.1 Quantum Algorithms - Grover's Search Algorithm

2.10.1.1 Quantum algorithm for Feynman loop integrals [120]

- **Authors:** Selomit Ramírez-Uribe, Andr  s E. Renter  a-Olivo, Germ  n Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

2.10.2 Quantum Algorithms - Quantum Walks

2.10.2.1 Quantum walk approach to simulating parton showers [121]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

2.10.3 Quantum Generative Adversarial Networks

2.10.3.1 Style-based quantum generative adversarial networks for Monte Carlo events [122]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

2.10.3.2 Quantum integration of elementary particle processes [123]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

2.10.3.3 Generative invertible quantum neural networks [124]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

2.10.3.4 Quantum-centric Supercomputing for Physics Research [102]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.10.4 Quantum Neural Networks

2.10.4.1 Loop Feynman integration on a quantum computer [125]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

2.10.5 Quantum Simulations

2.10.5.1 Quantum Algorithm for High Energy Physics Simulations [126]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

2.10.5.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [127]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

2.10.6 Variational Quantum Algorithms

2.10.6.1 Partonic collinear structure by quantum computing [128]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.10.6.2 Unsupervised quantum circuit learning in high energy physics [129]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.10.6.3 Quantum Wishlist: Lessons from Parton Showers [130]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in PoS:** 24 October 2025

2.10.7 Crosslists

2.10.7.1 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.10.7.2 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.10.7.3 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.10.7.4 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.10.7.5 Quantum data learning for quantum simulations in high-energy physics [131]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.10.7.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.10.7.7 Quantum Algorithms in Particle Physics [132]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.11 Fundamental Symmetries and Interactions

2.11.1 Quantum Entanglement and Bell Inequalities

2.11.1.1 Minimal entanglement and emergent symmetries in low-energy QCD [105]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.11.2 Crosslists

2.11.2.1 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.11.2.2 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.11.2.3 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Canelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

2.12 Gravitation and Astrophysics

2.12.1 Quantum Annealing

2.12.1.1 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [133]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

2.12.2 Quantum Sensors - Optical/Photonic Sensors

2.12.2.1 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [92]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.12.2.2 Electromagnetic cavities as mechanical bars for gravitational waves [93]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D’Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.12.2.3 Multimessenger Astronomy Beyond the Standard Model: New Window from Quantum Sensors [95]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025

2.12.3 Quantum Sensors - Optomechanical Sensors

2.12.3.1 Quantum interactions between a laser interferometer and gravitational waves [134]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

2.12.3.2 Superconducting Levitated Detector of Gravitational Waves [135]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

2.12.4 Crosslists

2.12.4.1 Laser Interferometers as Dark Matter Detectors [60]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

2.12.4.2 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Canelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.12.4.3 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.12.4.4 Search for composite dark matter with optically levitated sensors [34]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

2.12.4.5 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.12.4.6 Searching for Dark Matter with a Superconducting Qubit [62]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.12.4.7 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.12.4.8 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.12.4.9 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.12.4.10 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.12.4.11 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.12.4.12 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.12.4.13 Quantum measurements in fundamental physics: a user’s manual [64]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

2.13 Higgs Physics

2.13.1 Quantum Algorithms - Grover’s Search Algorithm

2.13.1.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [119]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.13.2 Quantum Annealing

2.13.2.1 Solving a Higgs optimization problem with quantum annealing for machine learning [114]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.13.2.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [115]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.13.3 Quantum Kernel Methods

2.13.3.1 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [117]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.13.3.2 Higgs analysis with quantum classifiers [118]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.13.4 Variational Quantum Algorithms

2.13.4.1 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [116]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.13.5 Crosslists

2.13.5.1 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.14 Jet Reconstruction

2.14.1 Quantum Algorithms - Grover's Search Algorithm

2.14.1.1 Quantum Algorithms for Jet Clustering [136]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.14.1.2 Quantum Algorithms in Particle Physics [132]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.14.2 Quantum Annealing

2.14.2.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [137]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

2.14.2.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [138]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

2.14.2.3 Quantum annealing for jet clustering with thrust [139]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.14.2.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [140]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

2.14.3 Quantum k-Means

2.14.3.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [141]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

2.14.3.2 Quantum clustering and jet reconstruction at the LHC [142]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

2.14.4 Tensor Networks

2.14.4.1 Quantum-inspired machine learning on high-energy physics data [143]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

2.14.4.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [144]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

2.14.4.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [145]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

2.14.5 Variational Quantum Algorithms

2.14.5.1 Quantum Machine Learning for b-jet charge identification [146]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

2.14.6 Crosslists

2.14.6.1 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.14.6.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.15 Lattice Scalar Theories

2.15.1 Quantum Simulations

2.15.1.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [147]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

2.15.1.2 Digitization of scalar fields for quantum computing [148]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

2.15.1.3 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [149]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

2.16 Lattice Fermion Theories

2.16.1 Quantum Simulations

2.16.1.1 Quantum Algorithms for Fermionic Quantum Field Theories [150]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

2.17 Lattice Gauge Theories

2.17.1 Quantum Annealing

2.17.1.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [151]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

2.17.1.2 SU(2) lattice gauge theory on a quantum annealer [152]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

2.17.2 Quantum Neural Networks

2.17.2.1 Quantum data learning for quantum simulations in high-energy physics [131]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.17.3 Quantum Simulations

2.17.3.1 Simulating lattice gauge theories on a quantum computer [153]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

2.17.3.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [154]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

2.17.3.3 Towards Quantum Simulating QCD [155]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

2.17.3.4 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [156]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

2.17.3.5 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [157]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

2.17.3.6 General Methods for Digital Quantum Simulation of Gauge Theories [158]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

2.17.3.7 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [159]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

2.17.3.8 Simulating Lattice Gauge Theories within Quantum Technologies [160]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

2.17.3.9 Toward scalable simulations of Lattice Gauge Theories on quantum computers [161]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

2.17.3.10 Role of boundary conditions in quantum computations of scattering observables [162]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

2.17.3.11 Quantum simulation of gauge theory via orbifold lattice [163]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

2.17.3.12 Lattice renormalization of quantum simulations [164]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

2.17.3.13 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [165]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

2.17.3.14 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [166]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

2.17.3.15 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [167]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

2.17.3.16 Measurement-based quantum simulation of Abelian lattice gauge theories [168]

- **Authors:** Hiroki Sukeno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

2.17.3.17 Quantum simulation of fundamental particles and forces [169]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

2.17.3.18 Toward QCD on quantum computer: orbifold lattice approach [170]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

2.17.3.19 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [171]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

2.17.3.20 Quantum Simulation of $SU(3)$ Lattice Yang-Mills Theory at Leading Order in Large- N_c Expansion [172]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

2.17.4 Variational Quantum Algorithms

2.17.4.1 $SU(2)$ hadrons on a quantum computer via a variational approach [173]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

2.17.5 Crosslists

2.17.5.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [174]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.17.5.2 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

2.17.5.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [7]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.17.5.4 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [10]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.17.5.5 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.17.5.6 Quantum computing hardware for HEP algorithms and sensing [15]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canceledo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor,

Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

2.17.5.7 Review on Quantum Computing for Lattice Field Theory [4]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

2.17.5.8 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.17.5.9 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [175]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.17.5.10 Steps toward quantum simulations of hadronization and energy loss in dense matter [176]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.18 Neutrinos

2.18.1 Quantum Sensors - Optomechanical Sensors

2.18.1.1 Searches for Massive Neutrinos with Mechanical Quantum Sensors [177]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

2.18.2 Variational Quantum Algorithms

2.18.2.1 Quantum convolutional neural networks for high energy physics data analysis [178]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.18.2.2 Hybrid Quantum-Classical Graph Convolutional Network [179]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.18.3 Crosslists

2.18.3.1 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum** 4, 027001, 2023: 01 May 2023

2.18.3.2 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.19 Nuclear Decays

2.19.1 Quantum Sensors - Optomechanical Sensors

2.19.1.1 Mechanical Detection of Nuclear Decays [180]

- **Authors:** Jiaxiang Wang, T.W. Penny, Juan Recoaro, Benjamin Siegel, Yu-Han Tseng, David C. Moore
- **Posted on arXiv:** 18 January 2024
- **Published in Phys. Rev. Lett.** 133, 023602 (2024): 08 July 2024

2.20 Quantum Chromodynamics

2.20.1 Quantum Entanglement and Bell Inequalities

2.20.1.1 Symmetry from entanglement suppression [181]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

2.20.2 Quantum Simulations

2.20.2.1 Quantum simulation of quantum field theory in the light-front formulation [182]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A** 105, 032418 (2022): 08 March 2022

2.20.2.2 Quantum simulation of colour in perturbative quantum chromodynamics [183]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

2.20.3 Crosslists

2.20.3.1 Partonic collinear structure by quantum computing [128]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.20.3.2 Minimal entanglement and emergent symmetries in low-energy QCD [105]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.21 Quantum Gravity

2.21.1 Crosslists

2.21.1.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell,

Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

2.21.1.2 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.21.1.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.21.1.4 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.21.1.5 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.22 Quantum Field Theories

2.22.1 Quantum Entanglement and Bell Inequalities

2.22.1.1 Flavor patterns of fundamental particles from quantum entanglement? [184]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.22.2 Crosslists

2.22.2.1 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.22.2.2 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [10]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.22.2.3 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.22.2.4 Loop Feynman integration on a quantum computer [125]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

2.22.2.5 Quantum Algorithms in Particle Physics [132]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.23 Spin Correlation in Particle Decays

2.23.1 Quantum Entanglement and Bell Inequalities

2.23.1.1 Entanglement and quantum tomography with top quarks at the LHC [185]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

2.23.1.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [186]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

2.23.1.3 Quantum tops at the LHC: from entanglement to Bell inequalities [187]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

2.23.1.4 Quantum SMEFT tomography: Top quark pair production at the LHC [104]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.23.1.5 Quantum information with top quarks in QCD [188]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

2.23.1.6 Improved tests of entanglement and Bell inequalities with LHC tops [189]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

2.23.1.7 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [190]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

2.23.1.8 Quantum Discord and Steering in Top Quarks at the LHC [191]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

2.23.1.9 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [192]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

2.23.1.10 Three-Body Entanglement in Particle Decays [193]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

2.23.1.11 Can Bell inequalities be tested via scattering cross-section at colliders ? [194]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

2.23.1.12 Quantum detection of new physics in top-quark pair production at the LHC [195]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
- **Published in JHEP:** 15 March 2024

2.23.1.13 Quantum entanglement and Bell inequality violation at colliders [196]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.23.1.14 Full quantum tomography of top quark decays [197]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

2.23.1.15 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [198]

- **Authors:** Yoav Afik, Federica Fabbri, Matthew Low, Luca Marzola, Juan Antonio Aguilar-Saavedra, Mohammad Mahdi Altakach, Nedaa Alexandra Asbah, Yang Bai, Hannah Banks, Alan J. Barr, Alexander Bernal, Thomas E. Browder, Paweł Caban, J. Alberto Casas, Kun Cheng, Frédéric Dèliot, Regina Demina, Antonio Di Domenico, Michał Eckstein, Marco Fabbrichesi, Benjamin Fuks, Emidio Gabrielli, Dorival Gonçalves, Radosław Grabarczyk, Michele Grossi, Tao Han, Timothy J. Hobbs, Paweł Horodecki, James Howarth, Shih-Chieh Hsu, Stephen Jiggins, Eleanor Jones, Andreas W. Jung, Andrea Helen Knue, Steffen Korn, Theodota Lagouri, Priyanka Lamba, Gabriel T. Landi, Haifeng Li, Qiang Li, Ian Low, Fabio Maltoni, Josh McFayden, Navin McGinnis, Roberto A. Morales, Jesús M. Moreno, Juan Ramón Muñoz de Nova, Giulia Negro, Davide Pagani, Giovanni Pelliccioli, Michele Pinamonti, Laura Pintucci, Baptiste Ravina, Alim Ruzi, Kazuki Sakurai, Ethan Simpson, Maximiliano Sioli, Shufang Su, Sokratis Trifinopoulos, Sven E. Vahsen, Sofia Vallecorsa, Alessandro Vicini, Marcel Vos, Eleni Vryonidou, Chris D. White, Martin J. White, Andrew J. Wildridge, Tong Arthur Wu, Laura Zani, Yulei Zhang, Knut Zoch
- **Posted on arXiv:** 31 March 2025
- **Published in Eur.Phys.J.Plus:** 09 September 2025

2.23.1.16 Colliders are Testing neither Locality via Bell’s Inequality nor Entanglement versus Non-Entanglement [199]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025

2.24 Statistical Analysis

2.24.1 Quantum Annealing

2.24.1.1 Unfolding measurement distributions via quantum annealing [200]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

2.24.1.2 A Quantum Algorithm for Model-Independent Searches for New Physics [24]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.25 Schwinger Model

2.25.1 Quantum Simulations

2.25.1.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [174]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.25.1.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [201]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

2.25.1.3 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [175]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.25.1.4 Steps toward quantum simulations of hadronization and energy loss in dense matter [176]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.25.2 Variational Quantum Algorithms

2.25.2.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [202]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

2.25.2.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [203]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

2.26 Track Reconstruction

2.26.1 Quantum Algorithms - Grover’s Search Algorithm

2.26.1.1 Quantum speedup for track reconstruction in particle accelerators [204]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

2.26.2 Quantum Algorithms - Harrow-Hassadim-Lloyd Algorithm

2.26.2.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [205]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pérez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

2.26.3 Quantum Annealing

2.26.3.1 A pattern recognition algorithm for quantum annealers [206]

- **Authors:** Frédéric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

2.26.3.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [207]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

2.26.3.3 Charged particle tracking with quantum annealing optimization [208]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

2.26.3.4 Quantum annealing algorithms for track pattern recognition [209]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

2.26.3.5 Particle track classification using quantum associative memory [210]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.26.3.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [211]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

2.26.4 Quantum Kernel Methods

2.26.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [212]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

2.26.5 Quantum Memory

2.26.5.1 Quantum Associative Memory in HEP Track Pattern Recognition [213]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

2.26.6 Quantum Neural Networks

2.26.6.1 Hybrid quantum classical graph neural networks for particle track reconstruction [214]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

2.26.7 Variational Quantum Algorithms

2.26.7.1 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [215]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

2.26.7.2 Particle track reconstruction with noisy intermediate-scale quantum computers [216]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023

2.26.7.3 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [217]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

2.26.8 Crosslists

2.26.8.1 Quantum convolutional neural networks for high energy physics data analysis [178]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.26.8.2 Hybrid Quantum-Classical Graph Convolutional Network [179]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.26.8.3 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.26.8.4 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.26.8.5 Quantum Systems for Enhanced High Energy Particle Physics Detectors [101]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.26.8.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3 Quantum Information Science in Nuclear and Particle Physics

3.1 Reviews and Whitepapers and Proceedings

3.1.1 Reviews

3.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics** **57 388 (2016):** 09 March 2016

3.1.1.2 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.1.1.3 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.1.1.4 Review on Quantum Computing for Lattice Field Theory [4]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

3.1.2 Whitepapers and Proceedings

3.1.2.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph

Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.1.2.2 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020

- **Published in Quantum Sci. Technol.** 6 024002, 2021 (invited): 18 January 2021

3.1.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [7]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma

- **Posted on arXiv:** 09 March 2022

3.1.2.4 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle

- **Posted on arXiv:** 14 March 2022

3.1.2.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.1.2.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [10]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.1.2.7 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.1.2.8 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatriwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz,

S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.1.2.9 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.1.2.10 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022

- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.1.2.11 Quantum computing hardware for HEP algorithms and sensing [15]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canceledo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

3.1.2.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.1.2.13 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.1.2.14 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.2 Continuous Variable Quantum Computing

3.2.1 Detector Technologies and Simulations

3.2.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [96]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.2.2 Crosslists

3.2.2.1 Unsupervised event classification with graphs on classical and photonic quantum computers [21]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.3 Quantum Algorithms - Grover's Search Algorithm

3.3.1 Beyond the Standard Model

3.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [25]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

3.3.2 Event Generation

3.3.2.1 Quantum algorithm for Feynman loop integrals [120]

- **Authors:** Selomit Ramírez-Urbe, Andr  s E. Renter  a-Olivo, Germ  n Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

3.3.3 Higgs Physics

3.3.3.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [119]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

3.3.4 Jet Reconstruction

3.3.4.1 Quantum Algorithms for Jet Clustering [136]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.3.4.2 Quantum Algorithms in Particle Physics [132]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.3.5 Track Reconstruction

3.3.5.1 Quantum speedup for track reconstruction in particle accelerators [204]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

3.3.6 Crosslists

3.3.6.1 Quantum clustering and jet reconstruction at the LHC [142]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.4 Quantum Algorithms - Harrow-Hassadim-Lloyd Algorithm

3.4.1 Track Reconstruction

3.4.1.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [205]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pèrez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

3.4.2 Crosslists

3.4.2.1 Quantum-centric Supercomputing for Physics Research [102]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.5 Quantum Algorithms - Quantum Walks

3.5.1 Event Generation

3.5.1.1 Quantum walk approach to simulating parton showers [121]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

3.6 Quantum Annealing

3.6.1 Detector Technologies and Simulations

3.6.1.1 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [97]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

3.6.2 Event Classification

3.6.2.1 Quantum algorithm for the classification of supersymmetric top quark events [28]

- **Authors:** Pedrame Bargassa, Timothée Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothée Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

3.6.2.2 Completely quantum neural networks [106]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

3.6.3 Gravitation and Astrophysics

3.6.3.1 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [133]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

3.6.4 Higgs Physics

3.6.4.1 Solving a Higgs optimization problem with quantum annealing for machine learning [114]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

3.6.4.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [115]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

3.6.5 Jet Reconstruction

3.6.5.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [137]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

3.6.5.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [138]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

3.6.5.3 Quantum annealing for jet clustering with thrust [139]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.6.5.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [140]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

3.6.6 Lattice Gauge Theories

3.6.6.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [151]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

3.6.6.2 SU(2) lattice gauge theory on a quantum annealer [152]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

3.6.7 Statistical Analysis

3.6.7.1 Unfolding measurement distributions via quantum annealing [200]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

3.6.7.2 A Quantum Algorithm for Model-Independent Searches for New Physics [24]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

3.6.8 Track Reconstruction

3.6.8.1 A pattern recognition algorithm for quantum annealers [206]

- **Authors:** Frédéric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

3.6.8.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [207]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

3.6.8.3 Charged particle tracking with quantum annealing optimization [208]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

3.6.8.4 Quantum annealing algorithms for track pattern recognition [209]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

3.6.8.5 Particle track classification using quantum associative memory [210]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.6.8.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [211]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

3.6.9 Crosslists

3.6.9.1 Quantum Algorithms for Jet Clustering [136]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.6.9.2 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.6.9.3 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.6.9.4 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.6.9.5 Preparations for quantum simulations of quantum chromodynamics in

$$1 + 1$$
 dimensions. I. Axial gauge [167]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemel'skiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.6.9.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.7 Quantum Entanglement and Bell Inequalities

3.7.1 Fundamental Symmetries and Interactions

3.7.1.1 Minimal entanglement and emergent symmetries in low-energy QCD [105]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

3.7.2 Quantum Chromodynamics

3.7.2.1 Symmetry from entanglement suppression [181]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

3.7.3 Quantum Field Theories

3.7.3.1 Flavor patterns of fundamental particles from quantum entanglement? [184]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.7.4 Spin Correlation in Particle Decays

3.7.4.1 Entanglement and quantum tomography with top quarks at the LHC [185]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.7.4.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [186]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

3.7.4.3 Quantum tops at the LHC: from entanglement to Bell inequalities [187]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

3.7.4.4 Quantum SMEFT tomography: Top quark pair production at the LHC [104]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

3.7.4.5 Quantum information with top quarks in QCD [188]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

3.7.4.6 Improved tests of entanglement and Bell inequalities with LHC tops [189]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

3.7.4.7 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [190]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

3.7.4.8 Quantum Discord and Steering in Top Quarks at the LHC [191]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

3.7.4.9 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [192]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

3.7.4.10 Three-Body Entanglement in Particle Decays [193]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

3.7.4.11 Can Bell inequalities be tested via scattering cross-section at colliders ? [194]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

3.7.4.12 Quantum detection of new physics in top-quark pair production at the LHC [195]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
- **Published in JHEP:** 15 March 2024

3.7.4.13 Quantum entanglement and Bell inequality violation at colliders [196]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.7.4.14 Full quantum tomography of top quark decays [197]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

3.7.4.15 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [198]

- **Authors:** Yoav Afik, Federica Fabbri, Matthew Low, Luca Marzola, Juan Antonio Aguilar-Saavedra, Mohammad Mahdi Altakach, Nedaa Alexandra Asbah, Yang Bai, Hannah Banks, Alan J. Barr, Alexander Bernal, Thomas E. Browder, Paweł Caban, J. Alberto Casas, Kun Cheng, Frédéric Dèliot, Regina Demina, Antonio Di Domenico, Michał Eckstein, Marco Fabbrichesi, Benjamin Fuks, Emidio Gabrielli, Dorival Gonçalves, Radosław Grabarczyk, Michele Grossi, Tao Han, Timothy J. Hobbs, Paweł Horodecki, James Howarth, Shih-Chieh Hsu, Stephen Jiggins, Eleanor Jones, Andreas W. Jung, Andrea Helen Knue, Steffen Korn, Theodota Lagouri, Priyanka Lamba, Gabriel T. Landi, Haifeng Li, Qiang Li, Ian Low, Fabio Maltoni, Josh McFayden, Navin McGinnis, Roberto A. Morales, Jesús M. Moreno, Juan Ramón Muñoz de Nova, Giulia Negro, Davide Pagani, Giovanni Pelliccioli, Michele Pinamonti, Laura Pintucci, Baptiste Ravina, Alim Ruzi, Kazuki Sakurai, Ethan Simpson, Maximiliano Sioli, Shufang Su, Sokratis Trifinopoulos, Sven E. Vahsen, Sofia Vallecorsa, Alessandro Vicini, Marcel Vos, Eleni Vryonidou, Chris D. White, Martin J. White, Andrew J. Wildridge, Tong Arthur Wu, Laura Zani, Yulei Zhang, Knut Zoch
- **Posted on arXiv:** 31 March 2025
- **Published in Eur.Phys.J.Plus:** 09 September 2025

3.7.4.16 Colliders are Testing neither Locality via Bell’s Inequality nor Entanglement versus Non-Entanglement [199]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025

3.7.5 Crosslists

3.7.5.1 Testing Bell Inequalities at the LHC with Top-Quark Pairs [186]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett.** **127 (2021) 16, 161801:** 15 October 2021

3.7.5.2 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.8 Quantum Generative Adversarial Networks

3.8.1 Detector Technologies and Simulations

3.8.1.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [98]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.8.1.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [99]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3.8.1.3 Precise image generation on current noisy quantum computing devices [100]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.8.2 Event Generation

3.8.2.1 Style-based quantum generative adversarial networks for Monte Carlo events [122]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

3.8.2.2 Quantum integration of elementary particle processes [123]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

3.8.2.3 Generative invertible quantum neural networks [124]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys.** 16, 146 (2024): 04 June 2024

3.8.2.4 Quantum-centric Supercomputing for Physics Research [102]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.8.3 Crosslists

3.8.3.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [96]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.8.3.2 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.8.3.3 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.9 Quantum Kernel Methods

3.9.1 Anomaly Detection

3.9.1.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [19]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

3.9.1.2 Quantum anomaly detection in the latent space of proton collision events at the LHC [20]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.9.2 Event Classification

3.9.2.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [107]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevier
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

3.9.2.2 Application of quantum machine learning in a Higgs physics study at the CEPC [108]

- **Authors:** Abdualazem Fadol, Qiyu Sha, Yaquan Fang, Zhan Li, Sitian Qian, Yuyang Xiao, Yu Zhang, Chen Zhou
- **Posted on arXiv:** 26 September 2022
- **Published in International Journal of Modern Physics A, 2024:** 07 March 2024

3.9.3 Higgs Physics

3.9.3.1 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [117]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research** **3**, 033221 (2021): 08 September 2021

3.9.3.2 Higgs analysis with quantum classifiers [118]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.9.4 Track Reconstruction

3.9.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [212]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

3.9.5 Crosslists

3.9.5.1 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.9.5.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.9.5.3 Quantum-centric Supercomputing for Physics Research [102]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.10 Quantum k-Means

3.10.1 Anomaly Detection

3.10.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [21]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.10.2 Jet Reconstruction

3.10.2.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [141]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.10.2.2 Quantum clustering and jet reconstruction at the LHC [142]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.10.3 Crosslists

3.10.3.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [20]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.11 Quantum Memory

3.11.1 Track Reconstruction

3.11.1.1 Quantum Associative Memory in HEP Track Pattern Recognition [213]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

3.11.2 Crosslists

3.11.2.1 Quantum clustering and jet reconstruction at the LHC [142]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.12 Quantum Neural Networks

3.12.1 Event Classification

3.12.1.1 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [109]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

3.12.1.2 Guided graph compression for quantum graph neural networks [110]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

3.12.2 Event Generation

3.12.2.1 Loop Feynman integration on a quantum computer [125]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

3.12.3 Lattice Gauge Theories

3.12.3.1 Quantum data learning for quantum simulations in high-energy physics [131]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.12.4 Track Reconstruction

3.12.4.1 Hybrid quantum classical graph neural networks for particle track reconstruction [214]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.12.5 Crosslists

3.12.5.1 Quantum convolutional neural networks for high energy physics data analysis [178]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.12.5.2 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [97]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

3.12.5.3 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.12.5.4 Quantum Algorithms in Particle Physics [132]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.13 Quantum Sensors - Atomic Sensors

3.13.1 Dark Matter - Particle-like Dark Matter

3.13.1.1 Trapped Electrons and Ions as Particle Detectors [30]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

3.13.1.2 Millicharged Dark Matter Detection with Ion Traps [31]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

3.13.2 Dark Matter - Wave-like Dark Matter

3.13.2.1 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [50]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

3.13.2.2 Quantum entanglement of ions for light dark matter detection [51]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

3.13.2.3 Highly excited electron cyclotron for QCD axion and dark-photon detection [52]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

3.13.2.4 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [53]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.13.2.5 Sensitively searching for microwave dark photons with atomic ensembles [54]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

3.13.2.6 Searches for exotic spin-dependent interactions with spin sensors [55]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.13.3 Detector Technologies and Simulations

3.13.3.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [101]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

3.13.3.2 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.13.4 Crosslists

3.13.4.1 Dark Matter Direct Detection with Accelerometers [73]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.13.4.2 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph

Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.13.4.3 Simulating Lattice Gauge Theories within Quantum Technologies [160]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.13.4.4 Intensity interferometry for ultralight bosonic dark matter detection [56]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.13.4.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.13.4.6 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.13.4.7 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.13.4.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.13.4.9 Detection of bosonovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer

- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.13.4.10 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.13.4.11 Multimessenger Astronomy Beyond the Standard Model: New Window from Quantum Sensors [95]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025

3.14 Quantum Sensors - Molecular Sensors

3.14.1 Crosslists

3.14.1.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell,

Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.14.1.2 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

3.14.1.3 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatriwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.14.1.4 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.14.1.5 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibbrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.15 Quantum Sensors - Nuclear Sensors

3.15.1 Dark Matter - Wave-like Dark Matter

3.15.1.1 Intensity interferometry for ultralight bosonic dark matter detection [56]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.15.1.2 Detection of bosonovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.15.1.3 A Nuclear Interferometer for Ultra-Light Dark Matter Detection [58]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024

3.15.1.4 Spin squeezing of macroscopic nuclear spin ensembles [59]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.15.2 Crosslists

3.15.2.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.15.2.2 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.15.2.3 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.15.2.4 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.15.2.5 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.16 Quantum Sensors - Optical/Photonic Sensors

3.16.1 Dark Matter - Wave-like Dark Matter

3.16.1.1 Laser Interferometers as Dark Matter Detectors [60]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

3.16.1.2 Accelerating dark-matter axion searches with quantum measurement technology [61]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

3.16.1.3 Searching for Dark Matter with a Superconducting Qubit [62]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

3.16.1.4 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [63]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

3.16.1.5 Quantum measurements in fundamental physics: a user’s manual [64]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

3.16.1.6 Quantum Enhancement in Dark Matter Detection with Quantum Computation [65]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

3.16.1.7 Search for QCD axion dark matter with transmon qubits and quantum circuit [66]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

3.16.1.8 Axion bounds from quantum technology [67]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

3.16.1.9 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [68]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

3.16.1.10 Transmon qubit modeling and characterization for Dark Matter search [69]

- **Authors:** R. Moretti, D. Labranca, P. Campana, R. Carobene, M. Gobbo, M.A. Castellanos-Beltran, D. Olaya, P.F. Hopkins, L. Banchi, M. Borghesi, A. Candido, S. Carrazza, H.A. Corti, A. D’Elia, M. Faverzani, E. Ferri, A. Nucciotti, L. Origo, A. Pasquale, A.S. Piedjou Komnang, A. Rettaroli, S. Tocci, C. Gatti, A. Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in nan:** 09 September 2024

3.16.1.11 Flux-Tunable Cavity for Dark Matter Detection [70]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

3.16.1.12 Probing sub-eV dark photon, scalar and axion-like particle dark matters with transmon qubits [71]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Published in Phys.Dark Univ.:** 15 November 2025

3.16.2 Gravitation and Astrophysics

3.16.2.1 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [92]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

3.16.2.2 Electromagnetic cavities as mechanical bars for gravitational waves [93]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D’Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

3.16.2.3 Multimessenger Astronomy Beyond the Standard Model: New Window from Quantum Sensors [95]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025

3.16.3 Crosslists

3.16.3.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.16.3.2 Simulating Lattice Gauge Theories within Quantum Technologies [160]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.16.3.3 Intensity interferometry for ultralight bosonic dark matter detection [56]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul

S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball

- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.16.3.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.16.3.5 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatriwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

3.16.3.6 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.16.3.7 Quantum computing hardware for HEP algorithms and sensing [15]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Canceledo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

3.16.3.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.16.3.9 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.16.3.10 Detection of bosonovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.16.3.11 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Canelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.16.3.12 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [53]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.17 Quantum Sensors - Optomechanical Sensors

3.17.1 Dark Matter - Particle-like Dark Matter

3.17.1.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [32]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

3.17.1.2 Proposal for gravitational direct detection of dark matter [33]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

3.17.1.3 Search for composite dark matter with optically levitated sensors [34]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

3.17.1.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [35]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

3.17.1.5 Requirements on quantum superpositions of macro-objects for sensing neutrinos [36]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

3.17.1.6 Optomechanical dark matter instrument for direct detection [37]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

3.17.1.7 Superfluid helium ultralight dark matter detector [38]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

3.17.1.8 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [39]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

3.17.1.9 Dark Matter Searches with Levitated Sensors [40]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

3.17.1.10 Probing Light Particles With Optically Trapped Sensors Through Nucleon Scattering [41]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025

3.17.1.11 Mechanical sensors for ultraheavy dark matter searches via long-range forces [42]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

3.17.2 Dark Matter - Wave-like Dark Matter

3.17.2.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [72]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

3.17.2.2 Dark Matter Direct Detection with Accelerometers [73]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.17.2.3 Ultralight dark matter detection with mechanical quantum sensors [74]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

3.17.2.4 Searching for scalar dark matter with compact mechanical resonators [75]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

3.17.2.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [76]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Communic.: 15 January 2021**

3.17.2.6 Searching for vector dark matter with an optomechanical accelerometer [77]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

3.17.2.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [78]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.: 01 September 2023**

3.17.2.8 Axion detection with optomechanical cavities [79]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B: 27 June 2024**

3.17.2.9 Vector wave dark matter and terrestrial quantum sensors [80]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP: 21 June 2024**

3.17.3 Dark Matter - Macroscopic Dark Matter

3.17.3.1 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [49]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.17.4 Gravitation and Astrophysics

3.17.4.1 Quantum interactions between a laser interferometer and gravitational waves [134]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

3.17.4.2 Superconducting Levitated Detector of Gravitational Waves [135]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

3.17.5 Neutrinos

3.17.5.1 Searches for Massive Neutrinos with Mechanical Quantum Sensors [177]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

3.17.6 Nuclear Decays

3.17.6.1 Mechanical Detection of Nuclear Decays [180]

- **Authors:** Jiaxiang Wang, T.W. Penny, Juan Recoaro, Benjamin Siegel, Yu-Han Tseng, David C. Moore
- **Posted on arXiv:** 18 January 2024
- **Published in Phys. Rev. Lett. 133, 023602 (2024):** 08 July 2024

3.17.7 Crosslists

3.17.7.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowering, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.17.7.2 Mechanical quantum sensing in the search for dark matter [6]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol.** 6 024002, 2021 (invited): 18 January 2021

3.17.7.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.17.7.4 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibbrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.17.7.5 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

3.17.7.6 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.17.7.7 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.17.7.8 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.17.7.9 Quantum measurements in fundamental physics: a user’s manual [64]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

3.18 Quantum Sensors - Solid State Sensors

3.18.1 Dark Matter - Particle-like Dark Matter

3.18.1.1 Dark matter direct detection with quantum dots [43]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

3.18.1.2 Dark Matter Induced Power in Quantum Devices [44]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

3.18.1.3 Transmon Qubit constraints on dark matter-nucleon scattering [45]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

3.18.1.4 Quantum Parity Detectors: a qubit based particle detection scheme with meV thresholds for rare-event searches [46]

- **Authors:** Karthik Ramanathan, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Brandon J. Sandoval, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024

3.18.1.5 Piezoelectric bulk acoustic resonators for dark photon detection [47]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

3.18.2 Dark Matter - Wave-like Dark Matter

3.18.2.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [81]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

3.18.2.2 Axion search with quantum nondemolition detection of magnons [82]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

3.18.2.3 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [83]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research** 4, 023162 (2022): 31 May 2022

3.18.2.4 Ultimate precision limit of noise sensing and dark matter search [84]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

3.18.2.5 Light dark matter search with nitrogen-vacancy centers in diamonds [85]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

3.18.2.6 Axion detection via superfluid ^3He ferromagnetic phase and quantum measurement techniques [86]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

3.18.2.7 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [87]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D** 111 (2025) 075028: 01 April 2025

3.18.2.8 Listening for new physics with quantum acoustics [88]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohriti Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

3.18.2.9 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [89]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

3.18.2.10 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [90]

- **Authors:** Wei Chao, Yu Gao, Ming-jie Jin, Xiao-sheng Liu, Xi-lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in nan:** 2025

3.18.2.11 Hybrid-spin decoupling for noise-resilient DC quantum sensing [91]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

3.18.3 Crosslists

3.18.3.1 Quantum Sensing for High Energy Physics [5]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph

Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O’Brien, Roger O’Brien, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.18.3.2 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [76]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.18.3.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [9]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.18.3.4 New Horizons: Scalar and Vector Ultralight Dark Matter [12]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M.

Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O’Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.18.3.5 Quantum Networks for High Energy Physics [13]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.18.3.6 Quantum Systems for Enhanced High Energy Particle Physics Detectors [101]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov

- **Published in nan:** 24 June 2022

3.18.3.7 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.18.3.8 Quantum sensing for particle physics [94]

- **Authors:** Steven D. Bass, Michael Doser

- **Posted on arXiv:** 19 May 2023

- **Published in Nature Rev.Phys.:** 16 April 2024

3.18.3.9 Quantum Sensors for High Energy Physics [18]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibbrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.18.3.10 Quantum measurements in fundamental physics: a user’s manual [64]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

3.18.3.11 Searches for exotic spin-dependent interactions with spin sensors [55]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.18.3.12 Spin squeezing of macroscopic nuclear spin ensembles [59]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.19 Quantum Simulations

3.19.1 Dark Matter - Particle-like Dark Matter

3.19.1.1 Quantum simulations of dark sector showers [48]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

3.19.2 Effective Field Theories

3.19.2.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [103]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

3.19.3 Event Generation

3.19.3.1 Quantum Algorithm for High Energy Physics Simulations [126]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

3.19.3.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [127]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

3.19.4 Lattice Scalar Theories

3.19.4.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [147]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

3.19.4.2 Digitization of scalar fields for quantum computing [148]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

3.19.4.3 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [149]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

3.19.5 Lattice Fermion Theories

3.19.5.1 Quantum Algorithms for Fermionic Quantum Field Theories [150]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

3.19.6 Lattice Gauge Theories

3.19.6.1 Simulating lattice gauge theories on a quantum computer [153]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

3.19.6.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [154]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

3.19.6.3 Towards Quantum Simulating QCD [155]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

3.19.6.4 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [156]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

3.19.6.5 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [157]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

3.19.6.6 General Methods for Digital Quantum Simulation of Gauge Theories [158]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

3.19.6.7 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [159]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

3.19.6.8 Simulating Lattice Gauge Theories within Quantum Technologies [160]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.19.6.9 Toward scalable simulations of Lattice Gauge Theories on quantum computers [161]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

3.19.6.10 Role of boundary conditions in quantum computations of scattering observables [162]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

3.19.6.11 Quantum simulation of gauge theory via orbifold lattice [163]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

3.19.6.12 Lattice renormalization of quantum simulations [164]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

3.19.6.13 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [165]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

3.19.6.14 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [166]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

3.19.6.15 Preparations for quantum simulations of quantum chromodynamics in $1+1$ dimensions. I. Axial gauge [167]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemel'skiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.19.6.16 Measurement-based quantum simulation of Abelian lattice gauge theories [168]

- **Authors:** Hiroki Sukeno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

3.19.6.17 Quantum simulation of fundamental particles and forces [169]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

3.19.6.18 Toward QCD on quantum computer: orbifold lattice approach [170]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

3.19.6.19 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [171]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.19.6.20 Quantum Simulation of $SU(3)$ Lattice Yang-Mills Theory at Leading Order in Large- N_c Expansion [172]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

3.19.7 Quantum Chromodynamics

3.19.7.1 Quantum simulation of quantum field theory in the light-front formulation [182]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

3.19.7.2 Quantum simulation of colour in perturbative quantum chromodynamics [183]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

3.19.8 Schwinger Model

3.19.8.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [174]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

3.19.8.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [201]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

3.19.8.3 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [175]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.19.8.4 Steps toward quantum simulations of hadronization and energy loss in dense matter [176]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.19.9 Crosslists

3.19.9.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.19.9.2 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [8]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.19.9.3 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [10]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Binnink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.19.9.4 Quantum Simulation for High-Energy Physics [14]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.19.9.5 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.19.9.6 Review on Quantum Computing for Lattice Field Theory [4]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

3.19.9.7 Quantum data learning for quantum simulations in high-energy physics [131]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.19.9.8 Loop Feynman integration on a quantum computer [125]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

3.20 Tensor Networks

3.20.1 Jet Reconstruction

3.20.1.1 Quantum-inspired machine learning on high-energy physics data [143]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

3.20.1.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [144]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

3.20.1.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [145]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

3.20.2 Crosslists

3.20.2.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.20.2.2 Simulating Lattice Gauge Theories within Quantum Technologies [160]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.20.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [7]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.20.2.4 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.20.2.5 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [16]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.20.2.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.21 Variational Quantum Algorithms

3.21.1 Anomaly Detection

3.21.1.1 Anomaly detection in high-energy physics using a quantum autoencoder [22]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.21.1.2 Quantum anomaly detection for collider physics [23]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

3.21.2 Event Classification

3.21.2.1 Event Classification with Quantum Machine Learning in High-Energy Physics [26]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

3.21.2.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [27]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

3.21.2.3 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [111]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

3.21.2.4 Quantum Metric Learning for New Physics Searches at the LHC [29]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

3.21.2.5 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [112]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

3.21.2.6 Quantum Vision Transformers for Quark–Gluon Classification [113]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

3.21.3 Event Generation

3.21.3.1 Partonic collinear structure by quantum computing [128]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.21.3.2 Unsupervised quantum circuit learning in high energy physics [129]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.21.3.3 Quantum Wishlist: Lessons from Parton Showers [130]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in PoS:** 24 October 2025

3.21.4 Higgs Physics

3.21.4.1 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [116]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

3.21.5 Jet Reconstruction

3.21.5.1 Quantum Machine Learning for b-jet charge identification [146]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

3.21.6 Lattice Gauge Theories

3.21.6.1 SU(2) hadrons on a quantum computer via a variational approach [173]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

3.21.7 Neutrinos

3.21.7.1 Quantum convolutional neural networks for high energy physics data analysis [178]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.21.7.2 Hybrid Quantum-Classical Graph Convolutional Network [179]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3.21.8 Schwinger Model

3.21.8.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [202]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

3.21.8.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [203]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

3.21.9 Track Reconstruction

3.21.9.1 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [215]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3.21.9.2 Particle track reconstruction with noisy intermediate-scale quantum computers [216]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023

3.21.9.3 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [217]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

3.21.10 Crosslists

3.21.10.1 Simulating Lattice Gauge Theories within Quantum Technologies [160]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.21.10.2 Quantum Machine Learning in High Energy Physics [2]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.21.10.3 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [98]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.21.10.4 Higgs analysis with quantum classifiers [118]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.21.10.5 Hybrid quantum classical graph neural networks for particle track reconstruction [214]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.21.10.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [10]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Binnink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borrás, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.21.10.7 Quantum Computing for Data Analysis in High-Energy Physics [11]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibeneditto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo

- **Posted on arXiv:** 15 March 2022

3.21.10.8 Quantum Computing Applications in Future Colliders [3]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.21.10.9 Quantum Computing for High-Energy Physics: State of the Art and Challenges [17]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borrás, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.21.10.10 Precise image generation on current noisy quantum computing devices [100]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.21.10.11 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [175]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.21.10.12 Quantum Algorithms in Particle Physics [132]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.21.10.13 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [171]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.21.10.14 Steps toward quantum simulations of hadronization and energy loss in dense matter [176]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

References

- [1] M. Dalmonte and S. Montangero, *Lattice gauge theory simulations in the quantum information era*, *Contemp. Phys.* **57** (2016) 388 [1602.03776].
- [2] W. Guan, G. Perdue, A. Pesah, M. Schuld, K. Terashi, S. Vallecorsa et al., *Quantum Machine Learning in High Energy Physics*, *Mach. Learn. Sci. Tech.* **2** (2021) 011003 [2005.08582].
- [3] H.M. Gray and K. Terashi, *Quantum Computing Applications in Future Colliders*, *Front. in Phys.* **10** (2022) 864823.
- [4] L. Funcke, T. Hartung, K. Jansen and S. Kühn, *Review on Quantum Computing for Lattice Field Theory*, *PoS LATTICE2022* (2023) 228 [2302.00467].
- [5] Z. Ahmed et al., *Quantum Sensing for High Energy Physics*, in *First workshop on Quantum Sensing for High Energy Physics*, 3, 2018 [1803.11306].
- [6] D. Carney et al., *Mechanical quantum sensing in the search for dark matter*, *Quantum Sci. Technol.* **6** (2021) 024002 [2008.06074].
- [7] Y. Meurice, J.C. Osborn, R. Sakai, J. Unmuth-Yockey, S. Catterall and R.D. Somma, *Tensor networks for High Energy Physics: contribution to Snowmass 2021*, in *Snowmass 2021*, 3, 2022 [2203.04902].
- [8] T. Faulkner, T. Hartman, M. Headrick, M. Rangamani and B. Swingle, *Snowmass white paper: Quantum information in quantum field theory and quantum gravity*, in *Snowmass 2021*, 3, 2022 [2203.07117].
- [9] O. Buchmueller et al., *Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks*, in *Snowmass 2021*, 3, 2022 [2203.07250].
- [10] T.S. Humble et al., *Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research*, in *Snowmass 2021*, 3, 2022 [2203.07091].
- [11] A. Delgado et al., *Quantum Computing for Data Analysis in High-Energy Physics*, in *Snowmass 2021*, 3, 2022 [2203.08805].
- [12] D. Antypas et al., *New Horizons: Scalar and Vector Ultralight Dark Matter*, 2203.14915.
- [13] A. Derevianko et al., *Quantum Networks for High Energy Physics*, in *Snowmass 2021*, 3, 2022 [2203.16979].
- [14] C.W. Bauer et al., *Quantum Simulation for High-Energy Physics*, *PRX Quantum* **4** (2023) 027001 [2204.03381].
- [15] M.S. Alam et al., *Quantum computing hardware for HEP algorithms and sensing*, in *Snowmass 2021*, 4, 2022 [2204.08605].
- [16] T.S. Humble, G.N. Perdue and M.J. Savage, *Snowmass Computational Frontier: Topical Group Report on Quantum Computing*, 2209.06786.
- [17] A. Di Meglio et al., *Quantum Computing for High-Energy Physics: State of the Art and Challenges*, *PRX Quantum* **5** (2024) 037001 [2307.03236].
- [18] A. Chou et al., *Quantum Sensors for High Energy Physics*, 11, 2023 [2311.01930].

- [19] J. Schuhmacher, L. Boggia, V. Belis, E. Puljak, M. Grossi, M. Pierini et al., *Unravelling physics beyond the standard model with classical and quantum anomaly detection*, *Mach. Learn. Sci. Tech.* **4** (2023) 045031 [2301.10787].
- [20] V. Belis, K.A. Woźniak, E. Puljak, P. Barkoutsos, G. Dissertori, M. Grossi et al., *Quantum anomaly detection in the latent space of proton collision events at the LHC*, *Commun. Phys.* **7** (2024) 334 [2301.10780].
- [21] A. Blance and M. Spannowsky, *Unsupervised event classification with graphs on classical and photonic quantum computers*, *JHEP* **21** (2020) 170 [2103.03897].
- [22] V.S. Ngairangbam, M. Spannowsky and M. Takeuchi, *Anomaly detection in high-energy physics using a quantum autoencoder*, *Phys. Rev. D* **105** (2022) 095004 [2112.04958].
- [23] S. Alvi, C.W. Bauer and B. Nachman, *Quantum anomaly detection for collider physics*, *JHEP* **02** (2023) 220 [2206.08391].
- [24] K.T. Matchev, P. Shyamsundar and J. Smolinsky, *A Quantum Algorithm for Model-Independent Searches for New Physics*, *LHEP* **2023** (2023) 301 [2003.02181].
- [25] A.E. Armenakas and O.K. Baker, *Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider*, 2010.00649.
- [26] K. Terashi, M. Kaneda, T. Kishimoto, M. Saito, R. Sawada and J. Tanaka, *Event Classification with Quantum Machine Learning in High-Energy Physics*, *Comput. Softw. Big Sci.* **5** (2021) 2 [2002.09935].
- [27] A. Blance and M. Spannowsky, *Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier*, *JHEP* **02** (2021) 212 [2010.07335].
- [28] P. Bargassa, T. Cabos, S. Cavinato, A. Cordeiro Oudot Choi and T. Hessel, *Quantum algorithm for the classification of supersymmetric top quark events*, *Phys. Rev. D* **104** (2021) 096004 [2106.00051].
- [29] A. Hammad, K. Kong, M. Park and S. Shim, *Quantum Metric Learning for New Physics Searches at the LHC*, 2311.16866.
- [30] D. Carney, H. Häffner, D.C. Moore and J.M. Taylor, *Trapped Electrons and Ions as Particle Detectors*, *Phys. Rev. Lett.* **127** (2021) 061804 [2104.05737].
- [31] D. Budker, P.W. Graham, H. Ramani, F. Schmidt-Kaler, C. Smorra and S. Ulmer, *Millicharged Dark Matter Detection with Ion Traps*, *PRX Quantum* **3** (2022) 010330 [2108.05283].
- [32] A. Kawasaki, *Search for Kilogram-scale Dark Matter with Precision Displacement Sensors*, *Phys. Rev. D* **99** (2019) 023005 [1809.00968].
- [33] D. Carney, S. Ghosh, G. Krnjaic and J.M. Taylor, *Proposal for gravitational direct detection of dark matter*, *Phys. Rev. D* **102** (2020) 072003 [1903.00492].
- [34] F. Monteiro, G. Afek, D. Carney, G. Krnjaic, J. Wang and D.C. Moore, *Search for composite dark matter with optically levitated sensors*, *Phys. Rev. Lett.* **125** (2020) 181102 [2007.12067].
- [35] G. Afek, D. Carney and D.C. Moore, *Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors*, *Phys. Rev. Lett.* **128** (2022) 101301 [2111.03597].

- [36] E. Kilian, M. Toroš, F.F. Deppisch, R. Saakyan and S. Bose, *Requirements on quantum superpositions of macro-objects for sensing neutrinos*, *Phys. Rev. Res.* **5** (2023) 023012 [2204.13095].
- [37] C.G. Baker, W.P. Bowen, P. Cox, M.J. Dolan, M. Goryachev and G. Harris, *Optomechanical dark matter instrument for direct detection*, *Phys. Rev. D* **110** (2024) 043005 [2306.09726].
- [38] M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh et al., *Superfluid helium ultralight dark matter detector*, *Phys. Rev. D* **109** (2024) 095011 [2309.07995].
- [39] X. Cheng, J.-H. Guo, W. Wang and B. Zhu, *Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment*, *Nucl. Phys. B* **1010** (2025) 116780 [2312.04202].
- [40] E. Kilian et al., *Dark Matter Searches with Levitated Sensors*, *AVS Quantum Sci.* **6** (2024) 030503 [2401.17990].
- [41] B. Dutta, D.K. Ghosh and S. Jeusun, *Probing Light Particles With Optically Trapped Sensors Through Nucleon Scattering*, 2502.00093.
- [42] J. Qin et al., *Mechanical sensors for ultraheavy dark matter searches via long-range forces*, *Phys. Rev. D* **112** (2025) 072003 [2503.11645].
- [43] C. Blanco, R. Essig, M. Fernandez-Serra, H. Ramani and O. Slone, *Dark matter direct detection with quantum dots*, *Phys. Rev. D* **107** (2023) 095035 [2208.05967].
- [44] A. Das, N. Kurinsky and R.K. Leane, *Dark Matter Induced Power in Quantum Devices*, *Phys. Rev. Lett.* **132** (2024) 121801 [2210.09313].
- [45] A. Das, N. Kurinsky and R.K. Leane, *Transmon Qubit constraints on dark matter-nucleon scattering*, *JHEP* **07** (2024) 233 [2405.00112].
- [46] K. Ramanathan, J.E. Parker, L.M. Joshi, A.D. Beyer, P.M. Echternach, S. Rosenblum et al., *Quantum Parity Detectors: a qubit based particle detection scheme with meV thresholds for rare-event searches*, 2405.17192.
- [47] T. Trickle, *Piezoelectric bulk acoustic resonators for dark photon detection*, *Phys. Rev. D* **112** (2025) 055043 [2501.05504].
- [48] S. Chigusa and M. Yamazaki, *Quantum simulations of dark sector showers*, *Phys. Lett. B* **834** (2022) 137466 [2204.12500].
- [49] C. Blanco, B. Elshimy, R.F. Lang and R. Orlando, *Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays*, *Phys. Rev. D* **105** (2022) 115031 [2112.14784].
- [50] X. Fan, G. Gabrielse, P.W. Graham, R. Harnik, T.G. Myers, H. Ramani et al., *One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector*, *Phys. Rev. Lett.* **129** (2022) 261801 [2208.06519].
- [51] A. Ito, R. Kitano, W. Nakano and R. Takai, *Quantum entanglement of ions for light dark matter detection*, *JHEP* **02** (2024) 124 [2311.11632].
- [52] X. Fan, G. Gabrielse, P.W. Graham, H. Ramani, S.S.Y. Wong and Y. Xiao, *Highly excited electron cyclotron for QCD axion and dark-photon detection*, *Phys. Rev. D* **111** (2025) 075022 [2410.05549].

- [53] J. Shu, B. Xu and Y. Xu, *Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection*, 2410.22413.
- [54] S. He, D. He, Y. Li, L. Gao, X. Feng, H. Zheng et al., *Sensitively searching for microwave dark photons with atomic ensembles*, *Phys. Rev. D* **112** (2025) 063536 [2412.00786].
- [55] M. Jiang, H. Su, Y. Chen, M. Jiao, Y. Huang, Y. Wang et al., *Searches for exotic spin-dependent interactions with spin sensors*, *Rept. Prog. Phys.* **88** (2025) 016401 [2412.03288].
- [56] H. Masia-Roig et al., *Intensity interferometry for ultralight bosonic dark matter detection*, *Phys. Rev. D* **108** (2023) 015003 [2202.02645].
- [57] J. Arakawa, J. Eby, M.S. Safronova, V. Takhistov and M.H. Zaheer, *Detection of bosonovae with quantum sensors on Earth and in space*, *Phys. Rev. D* **110** (2024) 075007 [2306.16468].
- [58] H. Banks, E. Fuchs and M. McCullough, *A Nuclear Interferometer for Ultra-Light Dark Matter Detection*, 2407.11112.
- [59] E. Boyers, G. Goldstein and A.O. Sushkov, *Spin squeezing of macroscopic nuclear spin ensembles*, *Phys. Rev. D* **111** (2025) 052004 [2502.14103].
- [60] E.D. Hall, R.X. Adhikari, V.V. Frolov, H. Müller, M. Pospelov and R.X. Adhikari, *Laser Interferometers as Dark Matter Detectors*, *Phys. Rev. D* **98** (2018) 083019 [1605.01103].
- [61] H. Zheng, M. Silveri, R.T. Brierley, S.M. Girvin and K.W. Lehnert, *Accelerating dark-matter axion searches with quantum measurement technology*, 1607.02529.
- [62] A.V. Dixit, S. Chakram, K. He, A. Agrawal, R.K. Naik, D.I. Schuster et al., *Searching for Dark Matter with a Superconducting Qubit*, *Phys. Rev. Lett.* **126** (2021) 141302 [2008.12231].
- [63] S. Chen, H. Fukuda, T. Inada, T. Moroi, T. Nitta and T. Sichanugrist, *Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits*, *Phys. Rev. Lett.* **131** (2023) 211001 [2212.03884].
- [64] J. Beceky, D. Carney and G. Marocco, *Quantum measurements in fundamental physics: a user's manual*, 2311.07270.
- [65] S. Chen, H. Fukuda, T. Inada, T. Moroi, T. Nitta and T. Sichanugrist, *Quantum Enhancement in Dark Matter Detection with Quantum Computation*, *Phys. Rev. Lett.* **133** (2024) 021801 [2311.10413].
- [66] S. Chen, H. Fukuda, T. Inada, T. Moroi, T. Nitta and T. Sichanugrist, *Search for QCD axion dark matter with transmon qubits and quantum circuit*, *Phys. Rev. D* **110** (2024) 115021 [2407.19755].
- [67] M. Bauer, S. Chakraborti and G. Rostagni, *Axion bounds from quantum technology*, *JHEP* **05** (2025) 023 [2408.06412].
- [68] H. Shi, A.J. Brady, W. Górecki, L. Maccone, R. Di Candia and Q. Zhuang, *Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse*, *npj Quantum Inf.* **11** (2025) 48 [2409.04656].
- [69] R. Moretti et al., *Transmon qubit modeling and characterization for Dark Matter search*, 2409.05988.

- [70] F. Zhao, Z. Li, A.V. Dixit, T. Roy, A. Vrajitoarea, R. Banerjee et al., *Flux-Tunable Cavity for Dark Matter Detection*, *Phys. Rev. Lett.* **135** (2025) 201002 [2501.06882].
- [71] W. Chao, Y. Gao, M.-J. Jin, X.-S. Liu and X.-L. Sun, *Probing sub-eV dark photon, scalar and axion-like particle dark matters with transmon qubits*, *Phys. Dark Univ.* **50** (2025) 102171.
- [72] A. Arvanitaki, S. Dimopoulos and K. Van Tilburg, *Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors*, *Phys. Rev. Lett.* **116** (2016) 031102 [1508.01798].
- [73] P.W. Graham, D.E. Kaplan, J. Mardon, S. Rajendran and W.A. Terrano, *Dark Matter Direct Detection with Accelerometers*, *Phys. Rev. D* **93** (2016) 075029 [1512.06165].
- [74] D. Carney, A. Hook, Z. Liu, J.M. Taylor and Y. Zhao, *Ultralight dark matter detection with mechanical quantum sensors*, *New J. Phys.* **23** (2021) 023041 [1908.04797].
- [75] J. Manley, R. Stump, D. Wilson, D. Grin and S. Singh, *Searching for scalar dark matter with compact mechanical resonators*, *Phys. Rev. Lett.* **124** (2020) 151301 [1910.07574].
- [76] L. Chen, J. Liu and K. Zhu, *Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum*, *Opt. Commun.* **479** (2021) 126422 [1911.12499].
- [77] J. Manley, M.D. Chowdhury, D. Grin, S. Singh and D.J. Wilson, *Searching for vector dark matter with an optomechanical accelerometer*, *Phys. Rev. Lett.* **126** (2021) 061301 [2007.04899].
- [78] A.J. Brady et al., *Entanglement-enhanced optomechanical sensor array with application to dark matter searches*, *Commun. Phys.* **6** (2023) 237 [2210.07291].
- [79] C. Murgui, Y. Wang and K.M. Zurek, *Axion detection with optomechanical cavities*, *Phys. Lett. B* **855** (2024) 138832 [2211.08432].
- [80] D.W.P. Amaral, M. Jain, M.A. Amin and C. Tunnell, *Vector wave dark matter and terrestrial quantum sensors*, *JCAP* **06** (2024) 050 [2403.02381].
- [81] X. Rong et al., *Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor*, *Nature Commun.* **9** (2018) 739 [1706.03482].
- [82] T. Ikeda, A. Ito, K. Miuchi, J. Soda, H. Kurashige and Y. Shikano, *Axion search with quantum nondemolition detection of magnons*, *Phys. Rev. D* **105** (2022) 102004 [2102.08764].
- [83] P.H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov et al., *Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors*, *Phys. Rev. Res.* **4** (2022) 023162 [2112.14882].
- [84] H. Shi and Q. Zhuang, *Ultimate precision limit of noise sensing and dark matter search*, *npj Quantum Inf.* **9** (2023) 27 [2208.13712].
- [85] S. Chigusa, M. Hazumi, E.D. Herbschleb, N. Mizuochi and K. Nakayama, *Light dark matter search with nitrogen-vacancy centers in diamonds*, *JHEP* **03** (2025) 083 [2302.12756].
- [86] S. Chigusa, D. Kondo, H. Murayama, R. Okabe and H. Sudo, *Axion detection via superfluid ^3He ferromagnetic phase and quantum measurement techniques*, *JHEP* **09** (2024) 191 [2309.09160].

- [87] S. Chigusa, M. Hazumi, E.D. Herbschleb, Y. Matsuzaki, N. Mizuochi and K. Nakayama, *Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection*, *Phys. Rev. D* **111** (2025) 075028 [2407.07141].
- [88] R. Linehan, T. Trickle, C.R. Conner, S. Ghosh, T. Lin, M. Sholapurkar et al., *Listening for new physics with quantum acoustics*, *Phys. Rev. D* **112** (2025) 115005 [2410.17308].
- [89] T. Sichanugrist, H. Fukuda, T. Moroi, K. Nakayama, S. Chigusa, N. Mizuochi et al., *Entanglement-enhanced ac magnetometry in the presence of Markovian noise*, *Phys. Rev. A* **111** (2025) 042605 [2410.21699].
- [90] W. Chao, Y. Gao, M.-j. Jin, X.-s. Liu and X.-l. Sun, *Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits*, *Phys. Dark Univ.* **50** (2025) 102171 [2412.20850].
- [91] S. Chigusa, M. Hazumi, E.D. Herbschleb, Y. Matsuzaki, N. Mizuochi and K. Nakayama, *Hybrid-spin decoupling for noise-resilient DC quantum sensing*, 2511.16732.
- [92] C. Dailey, C. Bradley, D.F. Jackson Kimball, I.A. Sulai, S. Pustelny, A. Wickenbrock et al., *Quantum sensor networks as exotic field telescopes for multi-messenger astronomy*, *Nature Astron.* **5** (2021) 150 [2002.04352].
- [93] A. Berlin, D. Blas, R. Tito D’Agnolo, S.A.R. Ellis, R. Harnik, Y. Kahn et al., *Electromagnetic cavities as mechanical bars for gravitational waves*, *Phys. Rev. D* **108** (2023) 084058 [2303.01518].
- [94] S.D. Bass and M. Doser, *Quantum sensing for particle physics*, *Nature Rev. Phys.* **6** (2024) 329 [2305.11518].
- [95] J. Arakawa, M.H. Zaheer, V. Takhistov, M.S. Safronova, J. Eby and C. Cheung, *Multimessenger Astronomy Beyond the Standard Model: New Window from Quantum Sensors*, 2502.08716.
- [96] S.Y. Chang, S. Vallecorsa, E.F. Combarro and F. Carminati, *Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors*, 2101.11132.
- [97] M. Schenk, E.F. Combarro, M. Grossi, V. Kain, K.S.B. Li, M.-M. Popa et al., *Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines*, *Quantum Sci. Technol.* **9** (2024) 025012 [2209.11044].
- [98] S.Y. Chang, S. Herbert, S. Vallecorsa, E.F. Combarro and R. Duncan, *Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics*, *EPJ Web Conf.* **251** (2021) 03050 [2103.15470].
- [99] S.Y. Chang, E. Agnew, E.F. Combarro, M. Grossi, S. Herbert and S. Vallecorsa, *Running the Dual-PQC GAN on noisy simulators and real quantum hardware*, *J. Phys. Conf. Ser.* **2438** (2023) 012062 [2205.15003].
- [100] F. Rehm, S. Vallecorsa, K. Borras, D. Krücker, M. Grossi and V. Varo, *Precise image generation on current noisy quantum computing devices*, *Quantum Sci. Technol.* **9** (2024) 015009 [2307.05253].
- [101] M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini et al., *Quantum Systems for Enhanced High Energy Particle Physics Detectors*, .

- [102] V.R. Pascuzzi and A. Córcoles, *Quantum-centric Supercomputing for Physics Research*, 8, 2024 [2408.11741].
- [103] C.W. Bauer, M. Freytsis and B. Nachman, *Simulating Collider Physics on Quantum Computers Using Effective Field Theories*, *Phys. Rev. Lett.* **127** (2021) 212001 [2102.05044].
- [104] R. Aoude, E. Madge, F. Maltoni and L. Mantani, *Quantum SMEFT tomography: Top quark pair production at the LHC*, *Phys. Rev. D* **106** (2022) 055007 [2203.05619].
- [105] Q. Liu, I. Low and T. Mehen, *Minimal entanglement and emergent symmetries in low-energy QCD*, *Phys. Rev. C* **107** (2023) 025204 [2210.12085].
- [106] S. Abel, J.C. Criado and M. Spannowsky, *Completely quantum neural networks*, *Phys. Rev. A* **106** (2022) 022601 [2202.11727].
- [107] J. Heredge, C. Hill, L. Hollenberg and M. Seviar, *Quantum Support Vector Machines for Continuum Suppression in B Meson Decays*, *Comput. Softw. Big Sci.* **5** (2021) 27 [2103.12257].
- [108] A. Fadol, Q. Sha, Y. Fang, Z. Li, S. Qian, Y. Xiao et al., *Application of quantum machine learning in a Higgs physics study at the CEPC*, *Int. J. Mod. Phys. A* **39** (2024) 2450007 [2209.12788].
- [109] Z. Li, L. Nagano and K. Terashi, *Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets*, *Phys. Rev. Res.* **6** (2024) 043028 [2405.11150].
- [110] M. Casals, V. Belis, E.F. Combarro, E. Alarcón, S. Vallecorsa and M. Grossi, *Guided graph compression for quantum graph neural networks*, *Mach. Learn. Sci. Tech.* **6** (2025) 035048 [2506.09862].
- [111] M.C. Peixoto, N.F. Castro, M. Crispim Romão, M.G.J. Oliveira and I. Ochoa, *Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets*, *Front. Artif. Intell.* **6** (2023) 1268852 [2211.03233].
- [112] E.B. Unlu et al., *Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics*, *Axioms* **13** (2024) 187 [2402.00776].
- [113] M.C. Cara et al., *Quantum Vision Transformers for Quark–Gluon Classification*, *Axioms* **13** (2024) 323 [2405.10284].
- [114] A. Mott, J. Job, J.R. Vlimant, D. Lidar and M. Spiropulu, *Solving a Higgs optimization problem with quantum annealing for machine learning*, *Nature* **550** (2017) 375.
- [115] A. Zlokapa, A. Mott, J. Job, J.-R. Vlimant, D. Lidar and M. Spiropulu, *Quantum adiabatic machine learning by zooming into a region of the energy surface*, *Phys. Rev. A* **102** (2020) 062405 [1908.04480].
- [116] S.L. Wu et al., *Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits*, *J. Phys. G* **48** (2021) 125003 [2012.11560].
- [117] S.L. Wu et al., *Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC*, *Phys. Rev. Res.* **3** (2021) 033221 [2104.05059].

- [118] V. Belis, S. González-Castillo, C. Reissel, S. Vallecorsa, E.F. Combarro, G. Dissertori et al., *Higgs analysis with quantum classifiers*, *EPJ Web Conf.* **251** (2021) 03070 [2104.07692].
- [119] A. Alexiades Armenakas and O.K. Baker, *Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider*, *Sci. Rep.* **11** (2021) 22850.
- [120] S. Ramírez-Urbe, A.E. Rentería-Olivo, G. Rodrigo, G.F.R. Sborlini and L. Vale Silva, *Quantum algorithm for Feynman loop integrals*, *JHEP* **05** (2022) 100 [2105.08703].
- [121] K. Bepari, S. Malik, M. Spannowsky and S. Williams, *Quantum walk approach to simulating parton showers*, *Phys. Rev. D* **106** (2022) 056002 [2109.13975].
- [122] C. Bravo-Prieto, J. Baglio, M. Cè, A. Francis, D.M. Grabowska and S. Carrazza, *Style-based quantum generative adversarial networks for Monte Carlo events*, *Quantum* **6** (2022) 777 [2110.06933].
- [123] G. Agliardi, M. Grossi, M. Pellen and E. Prati, *Quantum integration of elementary particle processes*, *Phys. Lett. B* **832** (2022) 137228 [2201.01547].
- [124] A. Rousselot and M. Spannowsky, *Generative invertible quantum neural networks*, *SciPost Phys.* **16** (2024) 146 [2302.12906].
- [125] J.J.M. de Lejarza, L. Cieri, M. Grossi, S. Vallecorsa and G. Rodrigo, *Loop Feynman integration on a quantum computer*, *Phys. Rev. D* **110** (2024) 074031 [2401.03023].
- [126] C.W. Bauer, W.A. de Jong, B. Nachman and D. Provasoli, *Quantum Algorithm for High Energy Physics Simulations*, *Phys. Rev. Lett.* **126** (2021) 062001 [1904.03196].
- [127] K. Bepari, S. Malik, M. Spannowsky and S. Williams, *Towards a quantum computing algorithm for helicity amplitudes and parton showers*, *Phys. Rev. D* **103** (2021) 076020 [2010.00046].
- [128] QUNU collaboration, *Partonic collinear structure by quantum computing*, *Phys. Rev. D* **105** (2022) L111502 [2106.03865].
- [129] A. Delgado and K.E. Hamilton, *Unsupervised quantum circuit learning in high energy physics*, *Phys. Rev. D* **106** (2022) 096006 [2203.03578].
- [130] M. Yamazaki, *Quantum Wishlist: Lessons from Parton Showers*, *PoS QCHSC24* (2025) 260 [2502.18059].
- [131] L. Nagano, A. Miessen, T. Onodera, I. Tavernelli, F. Tacchino and K. Terashi, *Quantum data learning for quantum simulations in high-energy physics*, *Phys. Rev. Res.* **5** (2023) 043250 [2306.17214].
- [132] G. Rodrigo, *Quantum Algorithms in Particle Physics*, *Acta Phys. Polon. Supp.* **17** (2024) 2 [2401.16208].
- [133] J. Caldeira, J. Job, S.H. Adachi, B. Nord and G.N. Perdue, *Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer*, 1911.06259.
- [134] B. Pang and Y. Chen, *Quantum interactions between a laser interferometer and gravitational waves*, *Phys. Rev. D* **98** (2018) 124006 [1808.09122].
- [135] D. Carney, G. Higgins, G. Marocco and M. Wentzel, *Superconducting Levitated Detector of Gravitational Waves*, *Phys. Rev. Lett.* **134** (2025) 181402 [2408.01583].

- [136] A.Y. Wei, P. Naik, A.W. Harrow and J. Thaler, *Quantum Algorithms for Jet Clustering*, *Phys. Rev. D* **101** (2020) 094015 [1908.08949].
- [137] D. Pires, Y. Omar and J. Seixas, *Adiabatic quantum algorithm for multijet clustering in high energy physics*, *Phys. Lett. B* **843** (2023) 138000 [2012.14514].
- [138] M. Kim, P. Ko, J.-h. Park and M. Park, *Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders*, 2111.07806.
- [139] A. Delgado and J. Thaler, *Quantum annealing for jet clustering with thrust*, *Phys. Rev. D* **106** (2022) 094016 [2205.02814].
- [140] E.R. Anschuetz, L. Funcke, P.T. Komiske, S. Kryhin and J. Thaler, *Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics*, *Phys. Rev. D* **106** (2022) 056008 [2205.10375].
- [141] D. Pires, P. Bargassa, J. Seixas and Y. Omar, *A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics*, 2101.05618.
- [142] J.J.M. de Lejarza, L. Cieri and G. Rodrigo, *Quantum clustering and jet reconstruction at the LHC*, *Phys. Rev. D* **106** (2022) 036021 [2204.06496].
- [143] T. Felser, M. Trenti, L. Sestini, A. Gianelle, D. Zuliani, D. Lucchesi et al., *Quantum-inspired machine learning on high-energy physics data*, *npj Quantum Inf.* **7** (2021) 111 [2004.13747].
- [144] J.Y. Araz and M. Spannowsky, *Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States*, *JHEP* **08** (2021) 112 [2106.08334].
- [145] J.Y. Araz and M. Spannowsky, *Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data*, *Phys. Rev. A* **106** (2022) 062423 [2202.10471].
- [146] A. Gianelle, P. Koppenburg, D. Lucchesi, D. Nicotra, E. Rodrigues, L. Sestini et al., *Quantum Machine Learning for b-jet charge identification*, *JHEP* **08** (2022) 014 [2202.13943].
- [147] S.P. Jordan, K.S.M. Lee and J. Preskill, *Quantum Computation of Scattering in Scalar Quantum Field Theories*, *Quant. Inf. Comput.* **14** (2014) 1014 [1112.4833].
- [148] N. Klco and M.J. Savage, *Digitization of scalar fields for quantum computing*, *Phys. Rev. A* **99** (2019) 052335 [1808.10378].
- [149] K. Yeter-Aydeniz, E.F. Dumitrescu, A.J. McCaskey, R.S. Bennink, R.C. Pooser and G. Siopsis, *Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers*, *Phys. Rev. A* **99** (2019) 032306 [1811.12332].
- [150] S.P. Jordan, K.S.M. Lee and J. Preskill, *Quantum Algorithms for Fermionic Quantum Field Theories*, 1404.7115.
- [151] N.T.T. Nguyen, G.T. Kenyon and B. Yoon, *A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer*, *Sci. Rep.* **10** (2020) 10915 [1911.06267].
- [152] S. A Rahman, R. Lewis, E. Mendicelli and S. Powell, *SU(2) lattice gauge theory on a quantum annealer*, *Phys. Rev. D* **104** (2021) 034501 [2103.08661].
- [153] T. Byrnes and Y. Yamamoto, *Simulating lattice gauge theories on a quantum computer*, *Phys. Rev. A* **73** (2006) 022328 [quant-ph/0510027].

- [154] U.-J. Wiese, *Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories*, *Annalen Phys.* **525** (2013) 777 [1305.1602].
- [155] U.-J. Wiese, *Towards Quantum Simulating QCD*, *Nucl. Phys. A* **931** (2014) 246 [1409.7414].
- [156] E. Zohar, J.I. Cirac and B. Reznik, *Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices*, *Rept. Prog. Phys.* **79** (2016) 014401 [1503.02312].
- [157] E.A. Martinez et al., *Real-time dynamics of lattice gauge theories with a few-qubit quantum computer*, *Nature* **534** (2016) 516 [1605.04570].
- [158] H. Lamm, S. Lawrence and Y. Yamauchi, *General Methods for Digital Quantum Simulation of Gauge Theories*, *Phys. Rev. D* **100** (2019) 034518 [1903.08807].
- [159] N. Klco, J.R. Stryker and M.J. Savage, *$SU(2)$ non-Abelian gauge field theory in one dimension on digital quantum computers*, *Phys. Rev. D* **101** (2020) 074512 [1908.06935].
- [160] M.C. Bañuls et al., *Simulating Lattice Gauge Theories within Quantum Technologies*, *Eur. Phys. J. D* **74** (2020) 165 [1911.00003].
- [161] S.V. Mathis, G. Mazzola and I. Tavernelli, *Toward scalable simulations of Lattice Gauge Theories on quantum computers*, *Phys. Rev. D* **102** (2020) 094501 [2005.10271].
- [162] R.A. Briceño, J.V. Guerrero, M.T. Hansen and A.M. Sturzu, *Role of boundary conditions in quantum computations of scattering observables*, *Phys. Rev. D* **103** (2021) 014506 [2007.01155].
- [163] A.J. Buser, H. Gharibyan, M. Hanada, M. Honda and J. Liu, *Quantum simulation of gauge theory via orbifold lattice*, *JHEP* **09** (2021) 034 [2011.06576].
- [164] M. Carena, H. Lamm, Y.-Y. Li and W. Liu, *Lattice renormalization of quantum simulations*, *Phys. Rev. D* **104** (2021) 094519 [2107.01166].
- [165] A. Kan and Y. Nam, *Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer*, 2107.12769.
- [166] C.W. Bauer and D.M. Grabowska, *Efficient representation for simulating $U(1)$ gauge theories on digital quantum computers at all values of the coupling*, *Phys. Rev. D* **107** (2023) L031503 [2111.08015].
- [167] R.C. Farrell, I.A. Chernyshev, S.J.M. Powell, N.A. Zemlevskiy, M. Illa and M.J. Savage, *Preparations for quantum simulations of quantum chromodynamics in 1+1 dimensions. I. Axial gauge*, *Phys. Rev. D* **107** (2023) 054512 [2207.01731].
- [168] H. Sukeno and T. Okuda, *Measurement-based quantum simulation of Abelian lattice gauge theories*, *SciPost Phys.* **14** (2023) 129 [2210.10908].
- [169] C.W. Bauer, Z. Davoudi, N. Klco and M.J. Savage, *Quantum simulation of fundamental particles and forces*, *Nature Rev. Phys.* **5** (2023) 420 [2404.06298].
- [170] G. Bergner, M. Hanada, E. Rinaldi and A. Schafer, *Toward QCD on quantum computer: orbifold lattice approach*, *JHEP* **05** (2024) 234 [2401.12045].
- [171] Z. Davoudi, C.-C. Hsieh and S.V. Kadam, *Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer*, *Quantum* **8** (2024) 1520 [2402.00840].

- [172] A.N. Ciavarella and C.W. Bauer, *Quantum Simulation of $SU(3)$ Lattice Yang-Mills Theory at Leading Order in Large- N_c Expansion*, *Phys. Rev. Lett.* **133** (2024) 111901 [2402.10265].
- [173] Y.Y. Atas, J. Zhang, R. Lewis, A. Jahanpour, J.F. Haase and C.A. Muschik, *$SU(2)$ hadrons on a quantum computer via a variational approach*, *Nature Commun.* **12** (2021) 6499 [2102.08920].
- [174] P. Hauke, D. Marcos, M. Dalmonte and P. Zoller, *Quantum simulation of a lattice Schwinger model in a chain of trapped ions*, *Phys. Rev. X* **3** (2013) 041018 [1306.2162].
- [175] R.C. Farrell, M. Illa, A.N. Ciavarella and M.J. Savage, *Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits*, *PRX Quantum* **5** (2024) 020315 [2308.04481].
- [176] R.C. Farrell, M. Illa and M.J. Savage, *Steps toward quantum simulations of hadronization and energy loss in dense matter*, *Phys. Rev. C* **111** (2025) 015202 [2405.06620].
- [177] D. Carney, K.G. Leach and D.C. Moore, *Searches for Massive Neutrinos with Mechanical Quantum Sensors*, *PRX Quantum* **4** (2023) 010315 [2207.05883].
- [178] S.Y.-C. Chen, T.-C. Wei, C. Zhang, H. Yu and S. Yoo, *Quantum convolutional neural networks for high energy physics data analysis*, *Phys. Rev. Res.* **4** (2022) 013231 [2012.12177].
- [179] S.Y.-C. Chen, T.-C. Wei, C. Zhang, H. Yu and S. Yoo, *Hybrid Quantum-Classical Graph Convolutional Network*, 2101.06189.
- [180] J. Wang, T.W. Penny, J. Recoaro, B. Siegel, Y.-H. Tseng and D.C. Moore, *Mechanical Detection of Nuclear Decays*, *Phys. Rev. Lett.* **133** (2024) 023602 [2402.13257].
- [181] I. Low and T. Mehen, *Symmetry from entanglement suppression*, *Phys. Rev. D* **104** (2021) 074014 [2104.10835].
- [182] M. Kreshchuk, W.M. Kirby, G. Goldstein, H. Beauchemin and P.J. Love, *Quantum simulation of quantum field theory in the light-front formulation*, *Phys. Rev. A* **105** (2022) 032418 [2002.04016].
- [183] H.A. Chawdhry and M. Pellen, *Quantum simulation of colour in perturbative quantum chromodynamics*, *SciPost Phys.* **15** (2023) 205 [2303.04818].
- [184] J. Thaler and S. Trifinopoulos, *Flavor patterns of fundamental particles from quantum entanglement?*, *Phys. Rev. D* **111** (2025) 056021 [2410.23343].
- [185] Y. Afik and J.R.M. de Nova, *Entanglement and quantum tomography with top quarks at the LHC*, *Eur. Phys. J. Plus* **136** (2021) 907 [2003.02280].
- [186] M. Fabbrichesi, R. Floreanini and G. Panizzo, *Testing Bell Inequalities at the LHC with Top-Quark Pairs*, *Phys. Rev. Lett.* **127** (2021) 161801 [2102.11883].
- [187] C. Severi, C.D.E. Boschi, F. Maltoni and M. Sioli, *Quantum tops at the LHC: from entanglement to Bell inequalities*, *Eur. Phys. J. C* **82** (2022) 285 [2110.10112].
- [188] Y. Afik and J.R.M. de Nova, *Quantum information with top quarks in QCD*, *Quantum* **6** (2022) 820 [2203.05582].
- [189] J.A. Aguilar-Saavedra and J.A. Casas, *Improved tests of entanglement and Bell inequalities with LHC tops*, *Eur. Phys. J. C* **82** (2022) 666 [2205.00542].

- [190] M. Fabbrichesi, R. Floreanini and E. Gabrielli, *Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs*, *Eur. Phys. J. C* **83** (2023) 162 [2208.11723].
- [191] Y. Afik and J.R.M. de Nova, *Quantum Discord and Steering in Top Quarks at the LHC*, *Phys. Rev. Lett.* **130** (2023) 221801 [2209.03969].
- [192] M. Fabbrichesi, R. Floreanini, E. Gabrielli and L. Marzola, *Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders*, *Eur. Phys. J. C* **83** (2023) 823 [2302.00683].
- [193] K. Sakurai and M. Spannowsky, *Three-Body Entanglement in Particle Decays*, *Phys. Rev. Lett.* **132** (2024) 151602 [2310.01477].
- [194] S. Li, W. Shen and J.M. Yang, *Can Bell inequalities be tested via scattering cross-section at colliders ?*, *Eur. Phys. J. C* **84** (2024) 1195 [2401.01162].
- [195] F. Maltoni, C. Severi, S. Tentori and E. Vryonidou, *Quantum detection of new physics in top-quark pair production at the LHC*, *JHEP* **03** (2024) 099 [2401.08751].
- [196] A.J. Barr, M. Fabbrichesi, R. Floreanini, E. Gabrielli and L. Marzola, *Quantum entanglement and Bell inequality violation at colliders*, *Prog. Part. Nucl. Phys.* **139** (2024) 104134 [2402.07972].
- [197] J.A. Aguilar-Saavedra, *Full quantum tomography of top quark decays*, *Phys. Lett. B* **855** (2024) 138849 [2402.14725].
- [198] Y. Afik et al., *Quantum information meets high-energy physics: input to the update of the European strategy for particle physics*, *Eur. Phys. J. Plus* **140** (2025) 855 [2504.00086].
- [199] S.A. Abel, H.K. Dreiner, R. Sengupta and L. Ubaldi, *Colliders are Testing neither Locality via Bell's Inequality nor Entanglement versus Non-Entanglement*, 2507.15949.
- [200] K. Cormier, R. Di Sipio and P. Wittek, *Unfolding measurement distributions via quantum annealing*, *JHEP* **11** (2019) 128 [1908.08519].
- [201] A.F. Shaw, P. Lougovski, J.R. Stryker and N. Wiebe, *Quantum Algorithms for Simulating the Lattice Schwinger Model*, *Quantum* **4** (2020) 306 [2002.11146].
- [202] N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz et al., *Quantum-classical computation of Schwinger model dynamics using quantum computers*, *Phys. Rev. A* **98** (2018) 032331 [1803.03326].
- [203] K. Ikeda, *Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning*, *Annalen Phys.* **537** (2025) 2400415 [2311.16297].
- [204] D. Magano et al., *Quantum speedup for track reconstruction in particle accelerators*, *Phys. Rev. D* **105** (2022) 076012 [2104.11583].
- [205] D. Nicotra, M. Lucio Martinez, J.A. de Vries, M. Merk, K. Driessens, R.L. Westra et al., *A quantum algorithm for track reconstruction in the LHCb vertex detector*, *JINST* **18** (2023) P11028 [2308.00619].
- [206] F. Bapst, W. Bhimji, P. Calafiura, H. Gray, W. Lavrijsen and L. Linder, *A pattern recognition algorithm for quantum annealers*, *Comput. Softw. Big Sci.* **4** (2020) 1 [1902.08324].
- [207] S. Das, A.J. Wildridge and A. Jung, *Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders*, 1903.08879.

- [208] A. Zlokapa, A. Anand, J.-R. Vlimant, J.M. Duarte, J. Job, D. Lidar et al., *Charged particle tracking with quantum annealing optimization*, *Quantum Machine Intelligence* **3** (2021) 27 [1908.04475].
- [209] M. Saito, P. Calafiura, H. Gray, W. Lavrijsen, L. Linder, Y. Okumura et al., *Quantum annealing algorithms for track pattern recognition*, *EPJ Web Conf.* **245** (2020) 10006.
- [210] G. Quiroz, L. Ice, A. Delgado and T.S. Humble, *Particle track classification using quantum associative memory*, *Nucl. Instrum. Meth. A* **1010** (2021) 165557 [2011.11848].
- [211] H. Okawa, Q.-G. Zeng, X.-Z. Tao and M.-H. Yung, *Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders*, *Comput. Softw. Big Sci.* **8** (2024) 16 [2402.14718].
- [212] P. Duckett, G. Facini, M. Jastrzebski, S. Malik, T. Scanlon and S. Rettie, *Reconstructing charged particle track segments with a quantum-enhanced support vector machine*, *Phys. Rev. D* **109** (2024) 052002 [2212.07279].
- [213] I. Shapoval and P. Calafiura, *Quantum Associative Memory in HEP Track Pattern Recognition*, *EPJ Web Conf.* **214** (2019) 01012 [1902.00498].
- [214] C. Tüysüz, C. Rieger, K. Novotny, B. Demirköz, D. Dobos, K. Potamianos et al., *Hybrid quantum classical graph neural networks for particle track reconstruction*, *Quantum Machine Intelligence* **3** (2021) 29 [2109.12636].
- [215] L. Funcke, T. Hartung, B. Heinemann, K. Jansen, A. Kropf, S. Kühn et al., *Studying quantum algorithms for particle track reconstruction in the LUXE experiment*, *J. Phys. Conf. Ser.* **2438** (2023) 012127 [2202.06874].
- [216] T. Schwägerl, C. Issever, K. Jansen, T.J. Khoo, S. Kühn, C. Tüysüz et al., *Particle track reconstruction with noisy intermediate-scale quantum computers*, 2303.13249.
- [217] H. Okawa, *Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm*, 2310.10255.